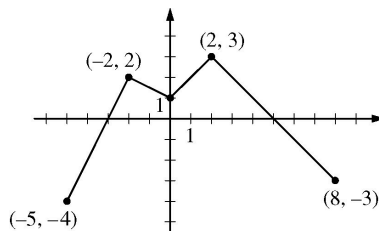


6-6

1.

Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) \, dt$.

Find $g(0)$ and $g(-5)$.

Part A

The response can earn up to 2 points:

1 point: For $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) \, dt = 3$$

1 point: For $g(-5)$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) \, dt = -7$$

✓

0	1	2
---	---	---

The response earns both of the following points:

1 point: For $g(0)$

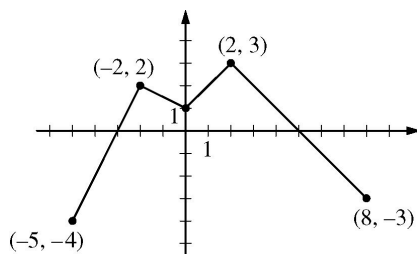
$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) \, dt = 3$$

1 point: For $g(-5)$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) \, dt = -7$$

6-6

2.

Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) dt$.

- Find $g(0)$ and $g(-5)$.
- Find $g'(x)$ in terms of $f(x)$. For each of $g'(4)$ and $g'(-2)$, find the value or state that it does not exist.
- On what intervals, if any, is the graph of g concave down? Give a reason for your answer.
- The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) dt = 3$$

1 point is earned for $g(-5)$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) dt = -10 + 3 = -7$$

✓

0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) dt = 3$$

1 point is earned for $g(-5)$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) dt = -10 + 3 = -7$$

Part B

The response can earn up to 3 points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

6-6

$g''(-2) = f'(-2)$ does not exist.

✓

0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$g''(-2) = f'(-2)$ does not exist.

Part C

The response can earn up to 1 point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

✓

0	1
---	---

The response earns all of the following point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

Part D

The response can earn up to 3 points:

1 point is earned for chain rule

$$h'(x) = g'(x^3 + 1) \cdot 3x^2$$

1 point is earned for correct answer

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$

✓

0	1	2	3
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The response earns all of the following point:

1 point is earned for chain rule

$$h'(x) = g'(x^3 + 1) \cdot 3x^2$$

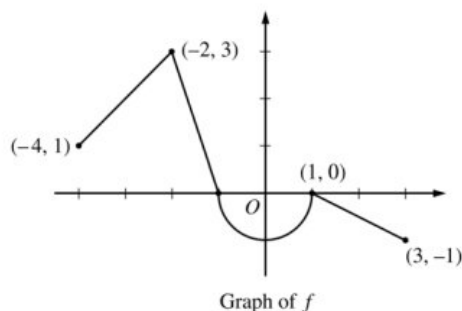
1 point is earned for correct answer

6-6

$$\begin{aligned}h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\&= (2 + 3) \cdot 3 = 15\end{aligned}$$

6-6

3.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

- Find the values of $g(2)$ and $g(-2)$.
- For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2} \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2} \end{aligned}$$

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

6-6

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

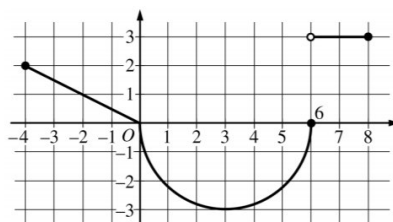
1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

6-6

6-6

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of g

The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above.

Let f be the function defined by $f(x) = 3x + \int_0^x g(t) dt$.

- (a) Find $f(7)$ and $f'(7)$.
- (b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.
- (c) For each of $\lim_{x \rightarrow 0^-} g'(x)$ and $\lim_{x \rightarrow 0^+} g'(x)$, find the value or state that it does not exist.
- (d) Find $\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1}$.

Part A

1 point is earned for: $f(7)$

1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $f(7)$

1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$

Part B

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

6-6

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

Part C

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$\lim_{x \rightarrow 0^+} g'(x)$ does not exist.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$\lim_{x \rightarrow 0^+} g'(x)$ does not exist.

Part D

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

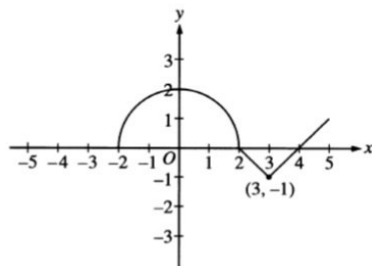
6-6

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

6-6

5.



The graph of the function f consists of a semicircle and two line segments as shown above. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

- Find $g(3)$.
- Find all the values of x on the open interval $(-2, 5)$ at which g has a relative maximum. Justify your answer.
- Write an equation for the line tangent to the graph of g at $x = 3$.
- Find the x -coordinate of each point of inflection of the graph of g on the open interval $(-2, 5)$. Justify your answer.

Part A

2 points are earned for the correct answer

<-1> each incorrect area

<-1> error in summing

$$\begin{aligned} g(3) &= \int_0^3 f(t) dt \\ &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2} \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the correct answer

<-1> each incorrect area

<-1> error in summing

$$\begin{aligned} g(3) &= \int_0^3 f(t) dt \\ &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2} \end{aligned}$$

Part B

1 point is earned for correctly answering that relative maximum at $x = 2$ only

1 point is earned for correctly answering $g'(x) = f(x)$ or interprets $g(x)$ as area accumulator

1 point is earned for the correct justification (ignore discussion at $x = 5$)

$g(x)$ has relative maximum at $x = 2$ because $g'(x) = f(x)$ changes from positive to negative at $x = 2$

✓

0	1	2	3
---	---	---	---

6-6

The student response earns all of the following points:

1 point is earned for correctly answering that relative maximum at $x = 2$ only

1 point is earned for correctly answering $g'(x) = f(x)$ or interprets $g(x)$ as area accumulator

1 point is earned for the correct justification (ignore discussion at $x = 5$)

$g(x)$ has relative maximum at $x = 2$ because $g'(x) = f(x)$ changes from positive to negative at $x = 2$

Part C

1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

$$y - \left(\pi - \frac{1}{2}\right) = -1(x - 3)$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

$$y - \left(\pi - \frac{1}{2}\right) = -1(x - 3)$$

Part D

1 point is earned for the correct points of inflection with x -coordinates 0 and 3 only

1 point is earned for the correct justification (ignore discussion at $x = 2$)

1/2 if $x = 0, 3$ selected as candidates and $x = 3$ discarded because $g''(3)$ does not exist

1/2 if $x = 0, 2, 3$ selected as candidates and $x = 2$ and $x = 3$ discarded because $g''(2)$ and $g''(3)$ do not exist

graph of g has points of inflection with x -coordinates $x = 0$ and $x = 3$

because $g''(x) = f(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = 3$

or

because $g'(x) = f(x)$ changes from increasing to decreasing at $x = 0$ and from decreasing to increasing at $x = 3$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct points of inflection with x -coordinates 0 and 3 only

6-6

1 point is earned for the correct justification (ignore discussion at $x = 2$)

1/2 if $x = 0$, 3 selected as candidates and $x = 3$ discarded because $g''(3)$ does not exist

1/2 if $x = 0$, 2, 3 selected as candidates and $x = 2$ and $x = 3$ discarded because $g''(2)$ and $g''(3)$ do not exist

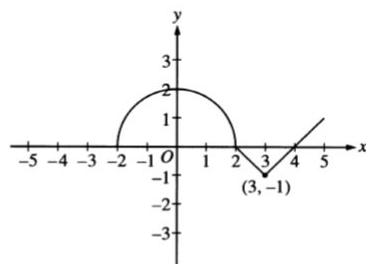
graph of g has points of inflection with x -coordinates $x = 0$ and $x = 3$

because $g''(x) = f(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = 3$

or

because $g'(x) = f(x)$ changes from increasing to decreasing at $x = 0$ and from decreasing to increasing at $x = 3$

6.



The graph of the function f consists of a semicircle and two line segments as shown above. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

Find $g(3)$.

Part A

2 points are earned for the correct answer

< -1 > each incorrect area

< -1 > error in summing

$$\begin{aligned} g(3) &= \int_0^3 f(t) dt \\ &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2} \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for the correct answer

< -1 > each incorrect area

< -1 > error in summing

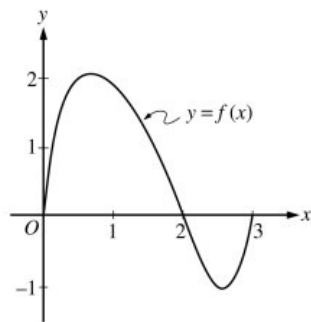
$$\begin{aligned} g(3) &= \int_0^3 f(t) dt \\ &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2} \end{aligned}$$

6-6

7. Let f and g be continuous functions such that $\int_0^{10} f(x)dx = 21$, $\int_0^{10} \frac{1}{2}g(x)dx = 8$, and $\int_3^{10} (f(x) - g(x))dx = 2$. What is the value of $\int_0^3 (f(x) - g(x))dx$?
- (A) 3 ✓
- (B) 7
- (C) 11
- (D) 15
- (E) 19
8. If $\int_4^{-10} g(x)dx = -3$ and $\int_4^6 g(x)dx = 5$, then $\int_{-10}^6 g(x)dx =$
- (A) -8
- (B) -2
- (C) 2
- (D) 8 ✓
9. Let f and g be continuous functions such that $\int_0^6 f(x)dx = 9$, $\int_3^6 f(x)dx = 5$, and $\int_3^0 g(x)dx = -7$. What is the value of $\int_0^3 \left(\frac{1}{2}f(x) - 3g(x)\right)dx$?
- (A) -23
- (B) -19 ✓
- (C) $-\frac{17}{2}$
- (D) 19
- (E) 23
10. Let f and g be continuous functions for $a \leq x \leq b$. If $a < c < b$, $\int_a^b f(x)dx = P$, $\int_c^b f(x)dx = Q$, $\int_a^b g(x)dx = R$, and $\int_c^b g(x)dx = S$, then $\int_a^c (f(x) - g(x))dx =$
- (A) $P - Q + R - S$
- (B) $P - Q - R + S$ ✓
- (C) $P - Q - R - S$
- (D) $P + Q - R - S$
- (E) $P + Q - R + S$

6-6

11.



The graph of a function f is shown above.

Which of the following expresses the relationship between $\int_0^2 f(x)dx$, $\int_0^3 f(x)dx$, and $\int_2^3 f(x)dx$?

(A) $\int_0^2 f(x)dx < \int_0^3 f(x)dx < \int_2^3 f(x)dx$

(B) $\int_0^3 f(x)dx < \int_0^2 f(x)dx < \int_2^3 f(x)dx$

(C) $\int_2^3 f(x)dx < \int_0^2 f(x)dx < \int_0^3 f(x)dx$

(D) $\int_2^3 f(x)dx < \int_0^3 f(x)dx < \int_0^2 f(x)dx$ ✓

12.

If $\int_0^1 f(x)dx = 2$ and $\int_0^4 f(x)dx = -3$, then $\int_1^4 (3f(x) + 2)dx =$

(A) -13

(B) -9 ✓

(C) -7

(D) 3

(E) 21

13.

If $\int_{-5}^2 f(x)dx = -17$ and $\int_5^2 f(x)dx = -4$, what is the value of $\int_{-5}^5 f(x)dx$?

(A) -21

(B) -13 ✓

(C) 0

(D) 13

(E) 21

14.

If $\int_0^3 f(x)dx = 6$ and $\int_3^5 f(x)dx = 4$, then $\int_0^5 (3 + 2f(x))dx =$

(A) 10

(B) 20

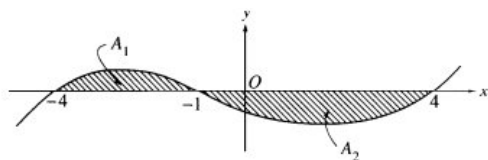
(C) 23

(D) 35 ✓

(E) 50

6-6

15.



The graph of $y=f(x)$ is shown in the figure above. If A_1 and A_2 are positive numbers that represent the areas of the shaded regions, then in terms of A_1 and A_2 ,

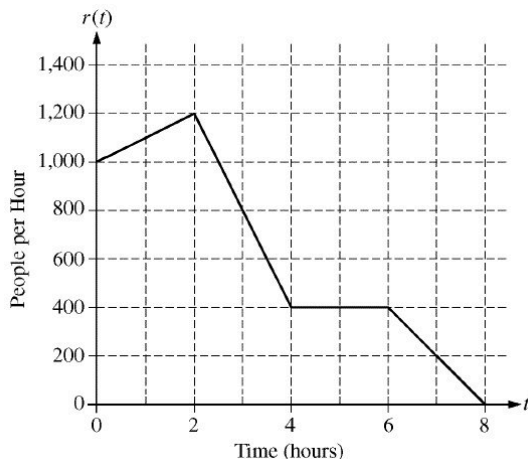
$$\int_{-4}^4 f(x)dx - 2 \int_{-1}^4 f(x)dx =$$

- (A) A_1
- (B) $A_1 - A_2$
- (C) $2A_1 - A_2$
- (D) $A_1 + A_2$
- (E) $A_1 + 2A_2$



6-6

16.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

- How many people arrive at the ride between $t = 0$ and $t = 3$? Show the computations that lead to your answer.
- Is the number of people waiting in line to get on the ride increasing or decreasing between $t = 2$ and $t = 3$? Justify your answer.
- At what time t is the line for the ride the longest? How many people are in line at that time? Justify your answers.
- Write, but do not solve, an equation involving an integral expression of r whose solution gives the earliest time t at which there is no longer a line for the ride.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

Part B

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

✓

0	1
---	---

The student response earns all of the following points:

6-6

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

Part C

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

Part D

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s) ds - 800t$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

6-6

$$0 = 700 + \int_0^t r(s)ds - 800t$$

17. The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

 The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t)dt + 11 \int_{17}^{23} E(t)dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t)dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was $15 * (6004) + 11 * (1271) = \$104,041$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t)dt + 11 \int_{17}^{23} E(t)dt = 104048.165$$

The amount collected was \$104,048.

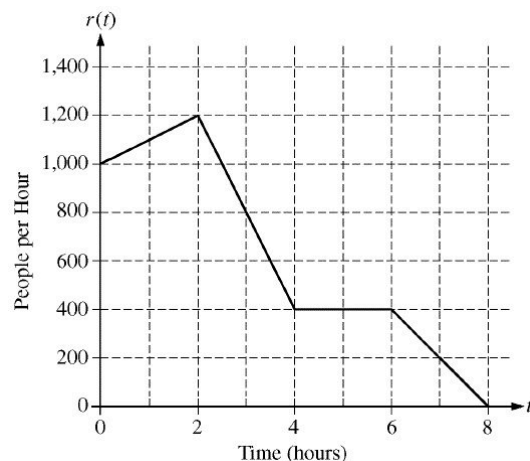
or

$$\int_{17}^{23} E(t)dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was $15 * (6004) + 11 * (1271) = \$104,041$.

6-6

18.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

How many people arrive at the ride between $t=0$ and $t=3$? Show the computations that lead to your answer.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$



0	1	2
---	---	---

The student response earns all of the following points:

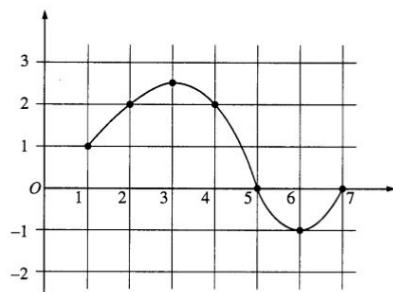
1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

6-6

19.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$ for $1 \leq x \leq 7$.

- Find $h(1)$.
- Find $h'(4)$.
- On what interval or intervals is the graph of h concave upward? Justify your answer.
- Find the value of x at which h has its minimum on the closed interval $[1, 7]$. Justify your answer.

Part A

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

Part B

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$

Part C

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$$1 < x < 3 \text{ and } 6 < x < 7$$

6-6

1 point is earned for the correct justification

h is concave up when:

- h' is increasing, or
- f is increasing, or
- $h''(x) > 0$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$1 < x < 3$ and $6 < x < 7$

1 point is earned for the correct justification

h is concave up when:

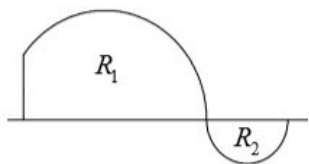
- h' is increasing, or
- f is increasing, or
- $h''(x) > 0$

Part D

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7 minimum at $x = 1$ because:

h increases on $[1,5]$ and decreases on $[5,7]$, so minimum is at an endpoint



$h(7) = \text{area } R_1 - \text{area } R_2 > 0$ and $h(1) = 0$

✓

0	1	2	3
---	---	---	---

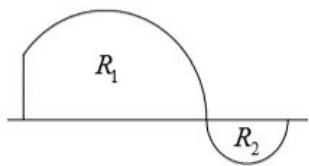
The student response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7 minimum at $x = 1$ because:

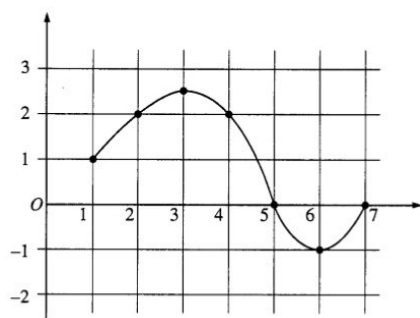
h increases on $[1,5]$ and decreases on $[5,7]$, so minimum is at an endpoint

6-6



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$

20.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$ for $1 \leq x \leq 7$.

Find $h(1)$.

Part A

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

✓

0

1

The student response earns one of the following points:

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

6-6

21.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

(a) Evaluate $\int_0^{1.5} (3f'(x) + 4)dx$. Show the work that leads to your answer.

(b) Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

(c) Find a positive real number r having the property that there must exist a value c with $0 < c < 0.5$ and $f''(c) = r$. Give a reason for your answer.

(d) Let g be the function given by $g(x) = \begin{cases} 2x^2 - x - 7 & \text{for } x < 0 \\ 2x^2 + x - 7 & \text{for } x \geq 0. \end{cases}$

The graph of g passes through each of the points $(x, f(x))$ given in the table above. Is it possible that f and g are the same function? Give a reason for your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4)dx &= 3 \int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4)dx &= 3 \int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$

Part B

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$\begin{aligned} y &= 5(x - 1) - 4 \\ f(1.2) &\approx 5(0.2) - 4 = -3 \end{aligned}$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$1 < x < 1.2$.

6-6



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$y = 5(x - 1) - 4$$

$$f(1.2) \approx 5(0.2) - 4 = -3$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$$1 < x < 1.2.$$

Part C

1 point is earned for the reference to MVT for f' (or differentiability of f)

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the reference to MVT for f' (or differentiability of f)

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$

Part D

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f is continuous at $x = 0$, so $f \neq g$.

6-6

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

22.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

Evaluate $\int_0^{1.5} (3f'(x) + 4) dx$. Show the work that leads to your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} f'(x) dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} f'(x) dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$

6-6

23. Let f be a function defined by $f(x) = \begin{cases} 1 - 2 \sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

- (a) Show that f is continuous at $x = 0$.
 (b) For $x \neq 0$, express $f'(x)$ as a piecewise-defined function. Find the value of x for which $f'(x) = -3$.
 (c) Find the average value of f on the interval $[-1, 1]$.

Part A

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.

Part B

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2 \cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$$-2 \cos x \neq -3 \text{ for all values for } x < 0.$$

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4} \ln \left(\frac{3}{4} \right) > 0.$$

6-6

Therefore, $f'(x) = -3$ for $x = -\frac{1}{4}\ln\left(\frac{3}{4}\right)$.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2\cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2\cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4}\ln\left(\frac{3}{4}\right) > 0.$$

Therefore, $f'(x) = -3$ for $x = -\frac{1}{4}\ln\left(\frac{3}{4}\right)$.

Part C

1 point is earned for $\int_{-1}^0 (1 - 2\sin x)dx$ and $\int_0^1 e^{-4x}dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x)dx &= \int_{-1}^0 f(x)dx + \int_0^1 f(x)dx \\ &= \int_{-1}^0 (1 - 2\sin x)dx + \int_0^1 e^{-4x}dx \\ &= [x + 2\cos]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2\cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x)dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4} \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

6-6

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned}\int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right)\end{aligned}$$

$$\begin{aligned}\text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4}\end{aligned}$$

6-6

24. Let f be a function defined by $f(x) = \begin{cases} 1 - 2 \sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

Find the average value of f on the interval $[-1, 1]$.

Part C

The student response earns none of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4} e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4} e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8} e^{-4} \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4} e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4} e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8} e^{-4} \end{aligned}$$

6-6

25. Let f and g have continuous first and second derivatives everywhere. If $f(x) \leq g(x)$ for all real x , which of the following must be true?

I. $f'(x) \leq g'(x)$ for all real x

II. $f''(x) \leq g''(x)$ for all real x .

III. $\int_0^1 f(x)dx \leq \int_0^1 g(x)dx$

(A) none

(B) I only

(C) III only

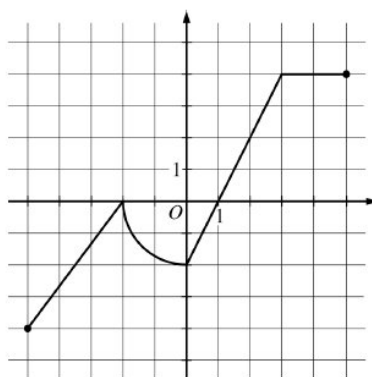
(D) I and II only

(E) I, II, and III



6-6

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of f

The graph of the function f , consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) dt$.

- Find the average rate of change of g from $x = -5$ to $x = 5$.
- Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.
- On what open intervals, if any, is the graph of g concave up? Justify your answer.
- Find all x -values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

Part A

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$$

Part B

1 point is earned for: answer

$$g'(x) = f(x)$$

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.

✓

0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer

6-6

$$g'(x) = f(x)$$

$$g'(3) = f(3) = 4$$

The instantaneous rate of change of g at $x = 3$ is 4.

Part C

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

Part D

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

6-6

27. The function f is defined by $f(x) = \begin{cases} 2 & \text{for } x < 3 \\ x - 1 & \text{for } x \geq 3. \end{cases}$ What is the value of $\int_1^5 f(x) dx$?

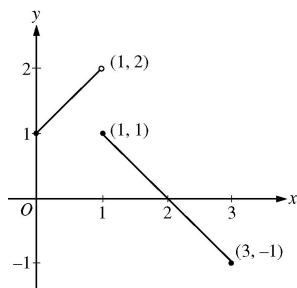
(A) 2
(B) 6
(C) 8

(D) 10

(E) 12



28.



Graph of f

The graph of the function f consists of two line segments, as shown in the figure above. The value of $\int_0^3 |f(x)| dx$ is

(A) $-\frac{3}{2}$

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) $\frac{5}{2}$

(E) nonexistent



6-6

29. Let f be the function defined by $f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$

Find the average value of $f(x)$ on the closed interval $0 \leq x \leq 5$.

Part B

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5-x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

$$\text{Average value: } \frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5-x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

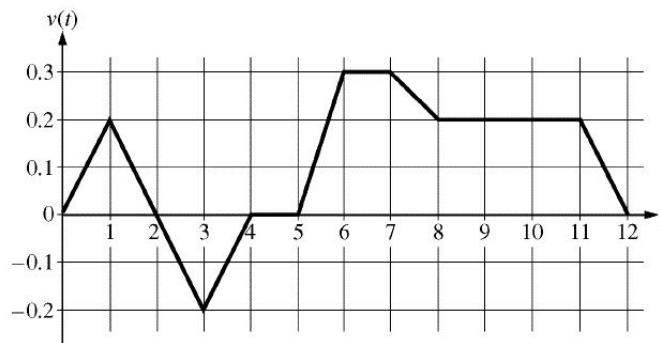
$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

$$\text{Average value: } \frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$$

6-6

30.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t = 0$ minutes and arriving at school at time $t = 12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

(a) Find the acceleration of Caren's bicycle at time $t = 7.5$ minutes. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_0^{12} |v(t)| dt$ in terms of Caren's trip. Find the value of $\int_0^{12} |v(t)| dt$.

(c) Shortly after leaving home, Caren realizes she left her calculus homework at home, and she returns to get it. At what time does she turn around to go back home? Give a reason for your answer.

(d) Larry also rides his bicycle along a straight road from home to school in 12 minutes. His velocity is modeled by the function w given by $w(t) = \frac{\pi}{15} \sin\left(\frac{\pi}{12}t\right)$, where $w(t)$ is in miles per minute for $0 \leq t \leq 12$ minutes. Who lives closer to school: Caren or Larry? Show the work that leads to your answer.

Part A

1 point is earned for the answer

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

Part B

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

6-6

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$

Part C

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative..

Part D

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$$\int_0^{12} w(t) dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

$$\int_0^{12} v(t) dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.

✓

0	1	2	3
---	---	---	---

The students response earns all of the following points:

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

6-6

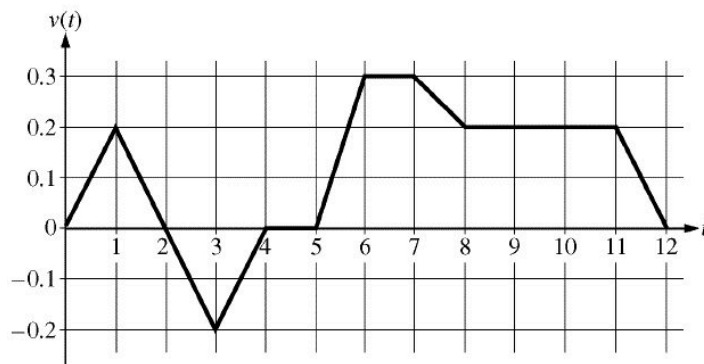
$$\int_0^{12} w(t)dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

$$\int_0^{12} v(t)dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.

6-6

31.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t=0$ minutes and arriving at school at time $t=12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

Using correct units, explain the meaning of $\int_0^{12} |v(t)| dt$ in terms of Caren's trip. Find the value of $\int_0^{12} |v(t)| dt$.

Part B

1 point is earned for meaning of integral

1 point is earned for value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t=0$ to $t=12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for meaning of integral

1 point is earned for value of integral

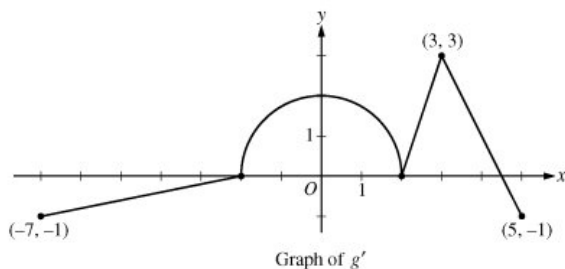
$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t=0$ to $t=12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$

6-6

32.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

Find $g(3)$ and $g(-2)$.

Part A

The student response earns none of the following points:

1 point is earned for using $g(0)=5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{4} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using $g(0)=5$

1 point is earned for $g(3)$

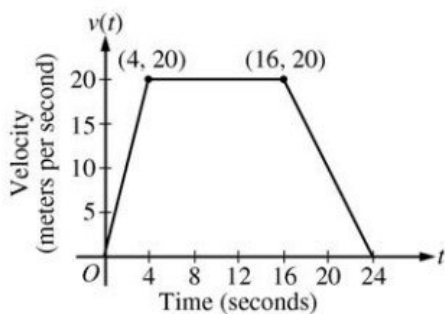
1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{4} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$

6-6

33.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

Find $\int_0^{24} v(t) dt$. Using correct units, explain the meaning of $\int_0^{24} v(t) dt$.

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

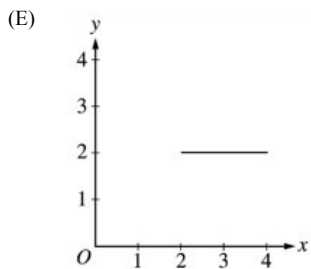
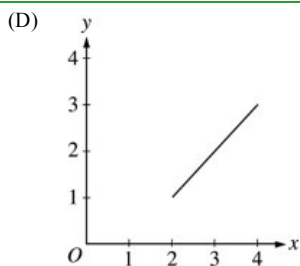
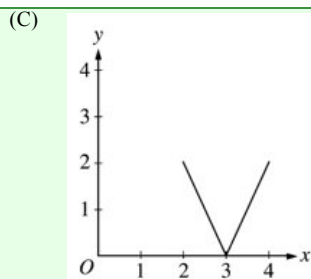
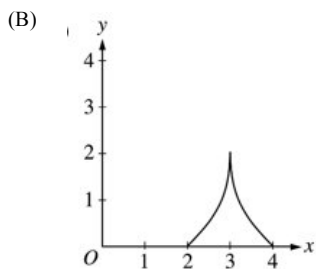
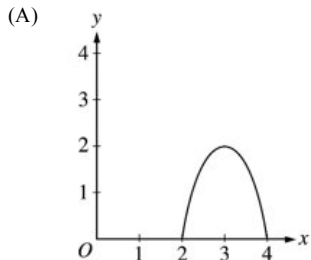
1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

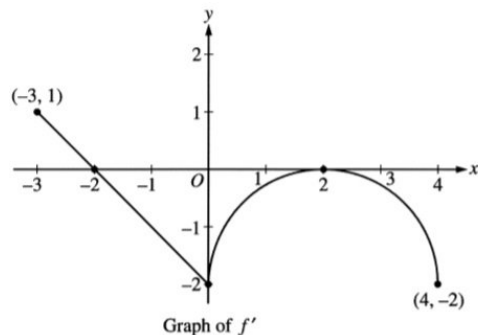
6-6

34. On the closed interval $[2, 4]$, which of the following could be the graph of a function f with the property that $\frac{1}{4-2} \int_2^4 f(t) dt = 1$?



6-6

35.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part D

1 point is earned for the difference of areas of triangles, $\pm (\frac{1}{2} - 2)$

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt = \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm (8 - \frac{1}{2}(2)^2\pi)$

$$f(4) - f(0) = \int_0^4 f'(t) dt = -(8 - \frac{1}{2}(2)^2\pi) = -8 + 2\pi$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the difference of areas of triangles, $\pm (\frac{1}{2} - 2)$

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt = \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm (8 - \frac{1}{2}(2)^2\pi)$

$$f(4) - f(0) = \int_0^4 f'(t) dt = -(8 - \frac{1}{2}(2)^2\pi) = -8 + 2\pi$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

6-6

36.

Let f be a continuous function on the closed interval $[0,2]$. If $2 \leq f(x) \leq 4$, then the greatest possible value of $\int_0^2 f(x)dx$ is

(A) 0

(B) 2

(C) 4

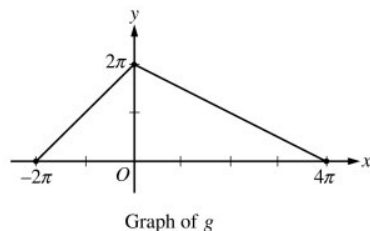
(D) 8

(E) 16



6-6

37.



Let g be the piecewise-linear function defined on $[-2\pi, 4\pi]$ whose graph is given above, and let $f(x) = g(x) - \cos\left(\frac{x}{2}\right)$.

- (a) Find $\int_{-2\pi}^{4\pi} f(x)dx$. Show the computations that lead to your answer.
- (b) Find all x -values in the open interval $(-2\pi, 4\pi)$ for which f has a critical point.
- (c) Let $h(x) = \int_0^{3x} g(t)dt$. Find $h'\left(-\frac{\pi}{3}\right)$.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\int_{-2\pi}^{4\pi} f(x)dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right)\right)dx \\ &= 6\pi^2 - \left[2\sin\left(\frac{x}{2}\right)\right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2\end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\int_{-2\pi}^{4\pi} f(x)dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right)\right)dx \\ &= 6\pi^2 - \left[2\sin\left(\frac{x}{2}\right)\right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2\end{aligned}$$

Part B

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

6-6

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

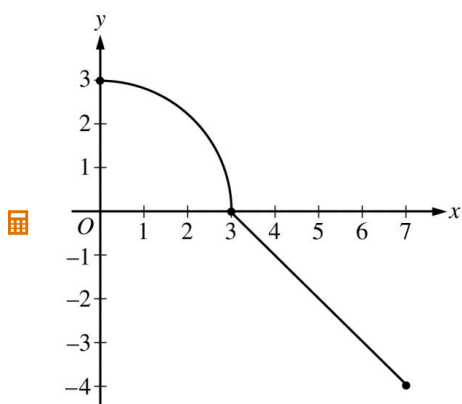
1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$

6-6

38.

Graph of f

The graph of the function f , which has a domain of $[0, 7]$, is shown in the figure above. The graph consists of a quarter circle of radius 3 and a segment with slope -1. Let b be a positive number such that $\int_0^b f(x) dx = 0$. What is the value of b ?

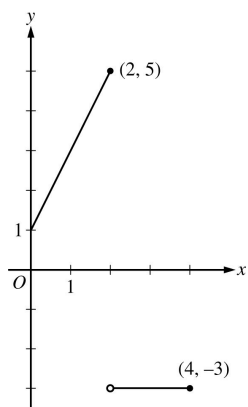
- (A) 3.760
 (B) 5.548
 (C) 5.659

(D) 6.760

(E) There is no such value of b .



39.

Graph of f

The graph of f is shown above for $0 \leq x \leq 4$. What is the value of $\int_0^4 f(x) dx$?

(A) -1

(B) 0

(C) 2

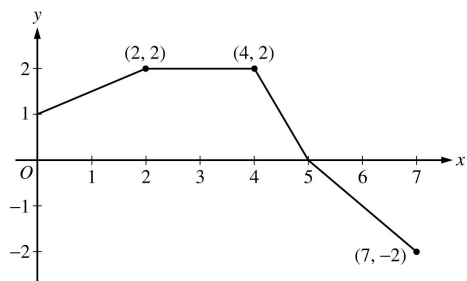
(D) 6

(E) 12



6-6

40.



The graph of a function f is shown above. What is the value of $\int_0^7 f(x) dx$?

(A) 6

(B) 8

(C) 10

(D) 14

(E) 18

41.

Given $f(x) = \begin{cases} x + 1 & \text{for } x < 0 \\ \cos \pi & \text{for } x \geq 0 \end{cases}$, then $\int_{-1}^1 f(x) dx =$

(A) $\frac{1}{2} + \frac{1}{\pi}$ (B) $-\frac{1}{2}$ (C) $\frac{1}{2} - \frac{1}{\pi}$ (D) $\frac{1}{2}$ (E) $-\frac{1}{2} + \pi$

42.

 If $f(x) = \begin{cases} x & \text{for } x \leq 1 \\ \frac{1}{x} & \text{for } x > 1, \end{cases}$ then $\int_0^e f(x) dx =$

(A) 0

(B) $\frac{3}{2}$

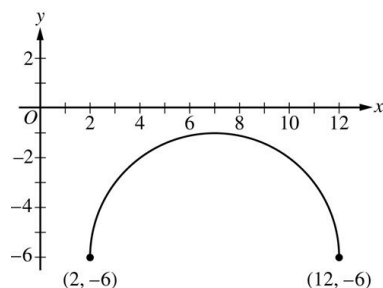
(C) 2

(D) e

(E) $e + \frac{1}{2}$

6-6

43.

Graph of f

The graph of $y = f(x)$ consists of a semicircle with endpoints at $(2, -6)$ and $(12, -6)$, as shown in the figure above. What is the value of $\int_2^{12} f(x) dx$

(A) $-\frac{25\pi}{2}$

(B) $\frac{25\pi}{2}$

(C) $-60 + \frac{25\pi}{2}$

(D) $60 - \frac{25\pi}{2}$

**Answer C**

This option is correct. The definite integral can be evaluated by using geometry and the connection of the definite integral with areas. In particular, the value of the integral here is the opposite of the area of the region between the semicircle and the x -axis, i.e. the area of the rectangle with vertices at $(2, 0)$, $(12, 0)$, $(2, -6)$, and $(12, -6)$ minus the area of the semicircle.

The area of the rectangle is $6 \cdot 10 = 60$. The area of semicircle of radius 5 is $\frac{1}{2}\pi(5)^2 = \frac{25\pi}{2}$. The definite integral therefore equal to $-(60 - \frac{25\pi}{2})$.

44.

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

The function f is defined above. The value of $\int_{-5}^3 f(x) dx$ is

(A) -2

(B) 2

(C) 8

(D) nonexistent

**Answer A**

This option is correct. This function has a jump discontinuity at $x = 0$. $f(x) = -1$ for $x < 0$ and $f(x) = 1$ for $x > 0$. Computing the definite integral in terms of areas and taking into account which region is below the axis and which is above, we get -2.

6-6

45.
$$f(x) = \begin{cases} \frac{|x|}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

The function f is defined above. The value of $\int_{-5}^3 f(x) dx$

(A) -2

(B) 2

(C) 8

(D) nonexistent

**Answer C**

This option is incorrect. Rather than taking into account the signed areas, this is total area:

$$\int_{-5}^3 f(x) dx = \text{area}(A) + \text{area}(B) = 5 + 3 = 8$$

or

$$\int_{-5}^3 f(x) dx = \left| \int_{-5}^0 -1 dx \right| + \left| \int_0^3 1 dx \right| = 5 + 3 = 8$$

46.
$$\int_{-1}^2 \frac{|x|}{x} dx$$
 is

(A) -3

(B) 1

(C) 2

(D) 3

(E) nonexistent



6-6

47. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function f defined by $f(t) = 2 \ln(t + 3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function g defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

 Part A

Find the rate of change of g at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let w be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim_{t \rightarrow 20^+} w(t)$. Is w continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function w defined in part C. Find the average value of w from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- P1** can also be earned for the exact value $\frac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$$g'(23) = -0.033$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(23)$

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- P2** is also earned for the units “liters/minute².”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.



0	1
---	---

The response accurately includes the criteria below.

6-6

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
 - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
 - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
 - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

$$30 + \int_0^{20} f(t)dt = 30 + \int_0^{20} (2 \ln(t + 3) - 1.6)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
 - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
 - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
 - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.



0	1
---	---

The response accurately includes the criteria below.

- Answer

6-6

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P5**, a response must indicate $\lim_{t \rightarrow 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
 - A response of “the right-hand limit is 4.704” is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \rightarrow 20^+} w(t) = \lim_{t \rightarrow 20^+} g(t) = 4.703562$$

The limit as t approaches 20 from the right is 4.704 (or 4.703).



0	1
---	---

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \rightarrow 20^-} w(t) = \lim_{t \rightarrow 20^-} f(t) = 4.670988$$

Because $\lim_{t \rightarrow 20^-} w(t) \neq \lim_{t \rightarrow 20^+} w(t)$, w is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of “ $\lim_{t \rightarrow 20^+} w(t) = 4.7$ and $\lim_{t \rightarrow 20^-} w(t) = w(20) = 4.7$, therefore continuous” earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, w is not continuous at $t = 20$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

6-6

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a , b , and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15} \int_{15}^{30} w(t)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
 - A response that earned **P7** earns **P8** for $\frac{1}{15} \int_{15}^{30} w(t)dt$, $\frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right)$, $\frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right)$, or equivalent.
 - Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work done in the calculator will be treated as scratch work and will not be considered in scoring.
 - “ $\int_{15}^{30} w(t)dt = 67.559$, so 4.504” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
 - “ $\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$ ” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.
 - A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15} \int_{15}^{30} f(t)dt$ or $\frac{1}{15} \int_{15}^{30} g(t)dt$.

Model Solution:

$$\begin{aligned} & \frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right) \\ &= \frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right) \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

6-6

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.



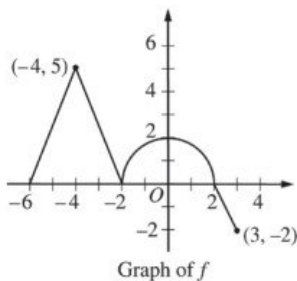
0

1

The response accurately includes the criteria below.

- Answer

48.



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by

$$g(x) = \int_{-2}^x f(t) dt.$$

Find $g'(0)$.

Part B

One point is earned for $g'(0)$

$$g'(0) = f(0) = 2$$



0

1

The student earns all of the following points:

One point is earned for $g'(0)$

$$g'(0) = f(0) = 2$$

49. $\int_1^4 |x - 3| dx =$

(A) $-\frac{3}{2}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{9}{2}$

(E) 5



50. If $\int_1^{10} f(x) dx = 4$ and $\int_{10}^3 f(x) dx = 7$ then $\int_1^3 f(x) dx =$

(A) -3

(B) 0

(C) 3

(D) 10

(E) 11



6-6

51.  If $\int_a^b f(x)dx = 5$ and $\int_a^b g(x)dx = -1$, which of the following must be true?

I. $f(x) > g(x)$ for $a \leq x \leq b$

II. $\int_a^b (f(x) + g(x))dx = 4$

III. $\int_a^b (f(x)g(x))dx = -5$

(A) I only

(B) II only

(C) III only

(D) II and III only

(E) I, II, and III

52. If f is a linear function and $0 < a < b$, then $\int_a^b f''(x)dx =$

(A) 0

(B) 1

(C) $\frac{ab}{2}$

(D) $b-a$

(E) $\frac{b^2-a^2}{2}$

53. What are all values of x for which $\int_x^2 t^3 dt$ is equal to 0?

(A) -2 only

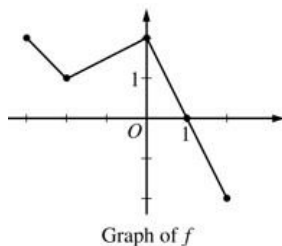
(B) 0 only

(C) 2 only

(D) -2 and 2 only

(E) -2, 0, and 2

- 54.



The graph of the piecewise linear function f is shown in the figure above. If $g(x) = \int_{-2}^x f(t)dt$, which of the following values is greatest?

(A) $g(-3)$

(B) $g(-2)$

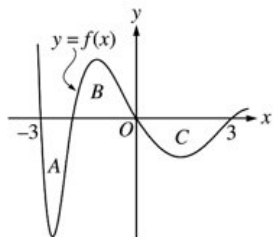
(C) $g(0)$

(D) $g(1)$

(E) $g(2)$

6-6

55.

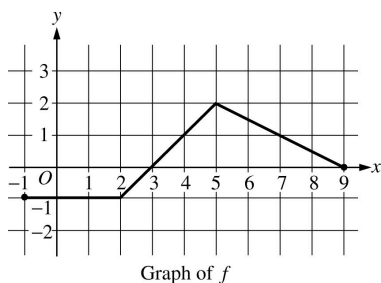


The regions A , B , and C in the figure above are bounded by the graph of the function f and the x -axis. If the area of each region is 2, what is the value of $\int_{-3}^3 (f(x) + 1) dx$?

- (A) -2
(B) -1
(C) 4
(D) 7
(E) 12



56.

Graph of f

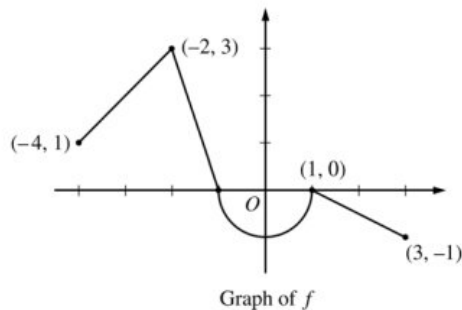
The graph of the piecewise linear function f is shown above. What is the value of $\int_{-1}^9 (3f(x) + 2) dx$?

- (A) 7.5
(B) 9.5
(C) 27.5
(D) 47
(E) 48.5



6-6

57.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

Find the values of $g(2)$ and $g(-2)$.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = - \int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $g(2)$

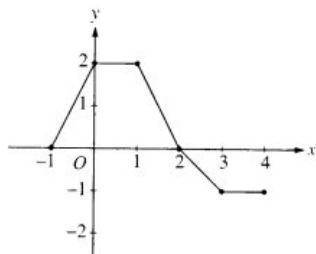
1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = - \int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$

6-6

58.



The graph of a piecewise-linear function f , for $-1 \leq x \leq 4$, is shown above. What is the value of $\int_{-1}^4 f(x) dx$?

(A) 1

(B) 2.5

(C) 4

(D) 5.5

(E) 8

59.

Let f and g be continuous functions. If $\int_2^6 f(x) dx = 5$ and $\int_6^2 g(x) dx = 7$, then $\int_2^6 (3f(x) + g(x)) dx =$

(A) -6

(B) 8

(C) 22

(D) 36

60.

Let f be a function such that $f(-x) = -f(x)$ for all x . If $\int_0^2 f(x) dx = 5$, then $\int_{-2}^2 (f(x) + 6) dx =$

(A) 6

(B) 16

(C) 24

(D) 34

61.

The function f is continuous on the closed interval $[1, 7]$. If $\int_1^7 f(x) dx = 42$ and $\int_7^3 f(x) dx = -32$ then $\int_1^3 2f(x) dx =$

(A) -148

(B) 10

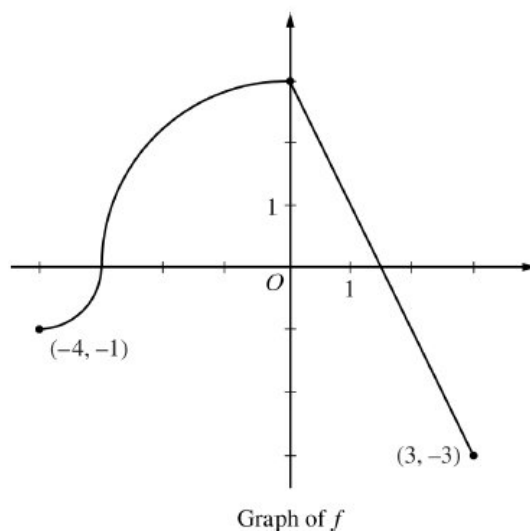
(C) 12

(D) 20

(E) 148

6-6

62.



The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

- (a) Find $g(-3)$. Find $g'(x)$ and evaluate $g'(-3)$.
- (b) Determine the x -coordinate of the point at which g has an absolute maximum on the interval $-4 \leq x \leq 3$. Justify your answer.
- (c) Find all values of x on the interval $-4 < x < 3$ for which the graph of g has a point of inflection. Give a reason for your answer.
- (d) Find the average rate of change of f on the interval $-4 \leq x \leq 3$. There is no point c , $-4 < c < 3$, for which $f'(c)$ is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part A

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t) dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

$$g'(-3) = 2 + f(-3) = 2$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t) dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

6-6

$$g'(-3) = 2 + f(-3) = 2$$

Part B

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.

Part C

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

6-6



0	1	2
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The student response earns all of the following points:

1 point is earned for average rate of change

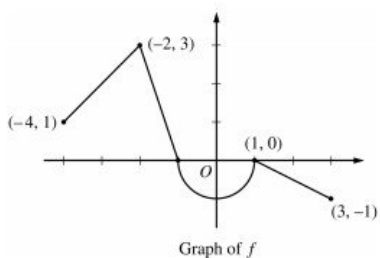
1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

6-6

63.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

- Find the values of $g(2)$ and $g(-2)$.
- For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

6-6

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

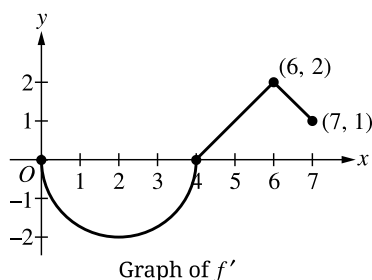
6-6

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be a differentiable function with $f(4) = 3$. On the interval $0 \leq x \leq 7$, the graph of f' , the derivative of f , consists of a semicircle and two line segments, as shown in the figure.

- Find $f(0)$ and $f(5)$.
- Find the x -coordinates of all points of inflection of the graph of f for $0 < x < 7$. Justify your answer.
- Let g be the function defined by $g(x) = f(x) - x$. On what intervals, if any, is g decreasing for $0 \leq x \leq 7$? Show the analysis that leads to your answer.
- For the function g defined in part (c), find the absolute minimum value on the interval $0 \leq x \leq 7$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with answers of only $f(0) = \pm 2\pi$, or only $f(5) = \frac{1}{2}$, or both earns 1 of the 3 points.

A response displaying $f(5) = \frac{7}{2}$ and a missing or incorrect value for $f(0)$ earns 2 of the 3 points.

The second and third points can be earned in either order.

Read unlabeled values from left to right and from top to bottom as $f(0)$ and $f(5)$. A single value must be labeled as either $f(0)$ or $f(5)$ in order to earn any points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Area of either region – OR – $\int_0^4 f'(x) dx$ – OR – $\int_4^5 f'(x) dx$
- ☐ $f(0)$
- ☐ $f(5)$

Solution:

$$f(0) = f(4) + \int_4^0 f'(x) dx = 3 - \int_0^4 f'(x) dx = 3 + 2\pi$$

6-6

$$f(5) = f(4) + \int_4^5 f'(x) dx = 3 + \frac{1}{2} = \frac{7}{2}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that gives only one of $x = 2$ or $x = 6$, along with a correct justification, earns 1 of the 2 points.

A response that claims that there is a point of inflection at any value other than $x = 2$ or $x = 6$ earns neither point.

To earn the second point a response must use correct reasoning based on the graph of f' . Examples of correct reasoning include:

- Correctly discussing the signs of the slopes of the graph of f'
- Citing $x = 2$ and $x = 6$ as the locations of local extrema on the graph of f'

Examples of reasoning not (sufficiently) connected to the graph of f' include:

- Reasoning based on sign changes in f'' unless the connection is made between the sign of f'' and the slopes of the graph of f'
- Reasoning based only on the concavity of the graph of f

The second point cannot be earned by use of vague or undefined terms such as “it” or “the function” or “the derivative.”

Responses that report inflection points as ordered pairs must report the points $(2, 3 + \pi)$ and $(6, 5)$ in order to earn the first point. If the y-coordinates are reported incorrectly, the response remains eligible for the second point.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

The graph of f has a point of inflection at each of $x = 2$ and $x = 6$, because $f'(x)$ changes from decreasing to increasing at $x = 2$ and from increasing to decreasing at $x = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned for $f'(x) \leq 1$ or the equivalent, in words or symbols.

Endpoints do not need to be included in the interval to be eligible for the second point.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(x) = f'(x) - 1$
- ☐ Interval with reason

Solution:

$$g'(x) = f'(x) - 1$$

$$f'(x) - 1 \leq 0 \Rightarrow f'(x) \leq 1$$

6-6

The graph of g is decreasing on the interval $0 \leq x \leq 5$ because $g'(x) \leq 0$ on this interval.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A justification that uses a local argument, such as “ g' changes from negative to positive (or g changes from decreasing to increasing) at $x = 5$ ” must also state that $x = 5$ is the only critical point.

If $g'(x) = 0$ (or equivalent) is not declared explicitly, a response that isolates $x = 5$ as the only critical number belonging to $(0, 7)$ earns the first point.

A response that imports $g'(x) = f'(x)$ from part (c) is eligible for the first point but not the second.

In this case, consideration of $x = 4$ as the only critical number on $(0, 7)$ earns the first point.

Solution using Candidates Test:

$$g'(x) = f'(x) - 1 = 0 \Rightarrow x = 5, x = 7$$

x	$g(x)$
0	$3 + 2\pi$
5	$-\frac{3}{2}$
7	$-\frac{1}{2}$

The absolute minimum value of g on the interval $0 \leq x \leq 7$ is $-\frac{3}{2}$.

When using a Candidates Test, a response may import an incorrect value of $f(0) = g(0) > -\frac{3}{2}$ from part (a). The second point can only be earned for an answer of $-\frac{3}{2}$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x) = 0$
- ☐ Answer with justification

Solution:

g is continuous, $g'(x) < 0$ for $0 < x < 5$, and $g'(x) > 0$ for $5 < x < 7$.

Therefore, the absolute minimum occurs at $x = 5$, and $g(5) = f(5) - 5 = \frac{7}{2} - 5 = -\frac{3}{2}$ is the absolute minimum value of g .

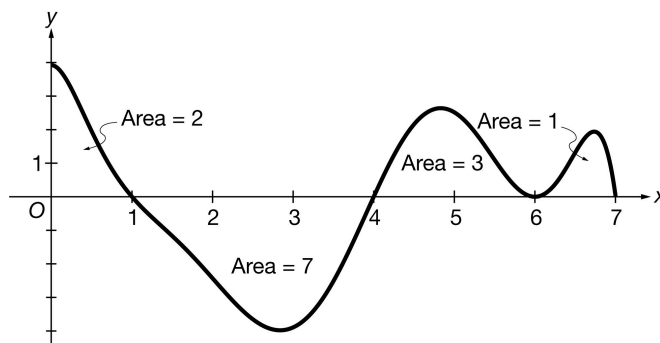
6-6

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a differentiable function f , on the closed interval $0 \leq x \leq 7$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(4) = 10$.

Let g be the function defined by $g(x) = 5 - x^2$.

- Find the value of $\int_0^7 f'(x) \, dx$.
- Given that $f(4) = 10$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of f and the absolute maximum value of f on the closed interval $0 \leq x \leq 7$. Justify your answers.
- Find $\int g(x) \, dx$.
- Find the value of $\int_1^2 x f'(g(x)) \, dx$.

Part A

The response should include supporting work for the numerical answer of -1 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^7 f'(x) \, dx = 2 - 7 + 3 + 1 = -1$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC in determining the appropriate $f(x)$ values for consideration.

For the second point: The response may also consider $x = 6$, though this is not required in order to justify an absolute minimum value or an absolute maximum value on the interval.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

6-6

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 1$ and $x = 4$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(4) + \int_4^x f'(t) dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$ or at an endpoint.

$$f'(x) = 0 \Rightarrow x = 1, x = 4, x = 6$$

The critical point at $x = 6$ is neither the location of an absolute minimum nor an absolute maximum because f' does not change sign at $x = 6$. Thus, the only candidates are $x = 0$, $x = 1$, $x = 4$, and $x = 7$.

x	$f(x)$
0	$f(4) + \int_4^0 f'(t) dt = f(4) - \int_0^4 f'(t) dt$ $= 10 - (2 - 7) = 15$
1	$f(4) + \int_4^1 f'(t) dt = f(4) - \int_1^4 f'(t) dt$ $= 10 - (-7) = 17$
4	10
7	$f(4) + \int_4^7 f'(t) dt = 10 + (3 + 1) = 14$

On the interval $0 \leq x \leq 7$, the absolute minimum value is $f(4) = 10$ and the absolute maximum value is $f(1) = 17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer.

Solution:

$$\int g(x) dx = \int (5 - x^2) dx = 5x - \frac{1}{3}x^3 + C$$

Part D

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-6



0	1	2	3
---	---	---	---

The student accurately includes all three of the criteria below.

- ☐ sets $u = 5 - x^2$
- ☐ integral in terms of u
- ☐ answer

Solution:

Let $u = g(x) = 5 - x^2$.

Then $du = -2x \, dx$ and $-\frac{1}{2} du = x \, dx$.

$x = 2 \Rightarrow u = 5 - 2^2 = 1$

$x = 1 \Rightarrow u = 5 - 1^2 = 4$

$$\int_1^2 x f'(g(x)) \, dx = -\frac{1}{2} \int_4^1 f'(u) \, du = \frac{1}{2} \int_1^4 f'(u) \, du = \frac{1}{2}(-7) = -\frac{7}{2}$$

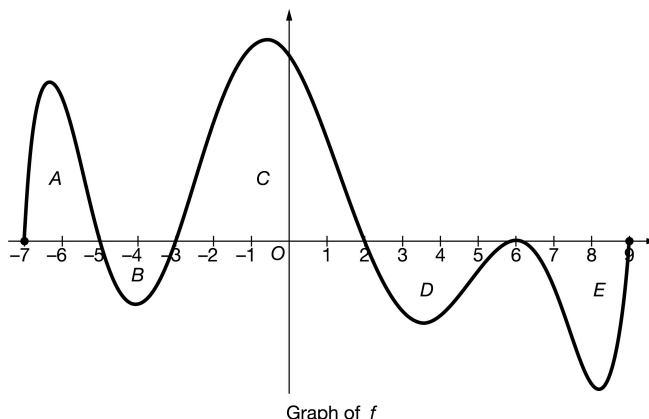
6-6

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The figure above shows the graph of the continuous function f . The regions A , B , C , D , and E have areas 5, 2, 16, 5, and 6, respectively. For $-7 \leq x \leq 9$, the function g is defined by $g(x) = -6 + \int_{-3}^x f(t) \, dt$.

- Is there a value of x , for $-3 \leq x \leq 2$, such that $g(x) = 0$? Justify your answer.
- Find the absolute minimum value of g and the absolute maximum value of g on the interval $-7 \leq x \leq 9$. Justify your answer.
- Find $\int (2x + 16) \, dx$.
 - Find the value of $\int_{-7}^{-5} f(2x + 16) \, dx$.

Part A

For the second point, the response must include that g is continuous on $-3 \leq x \leq 2$, show that $g(-3) < 0 < g(2)$, and draw the conclusion from the Intermediate Value Theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2
---	---	---



The student response accurately includes both of the criteria below.

- ☐ $g(-3)$ and $g(2)$
- ☐ justification, using Intermediate Value Theorem

Solution:

$g'(x) = f(x)$ and f is continuous on $-7 \leq x \leq 9$. Because g is an antiderivative of a continuous function f , g is continuous on $-3 \leq x \leq 2$.

$$g(-3) = -6 + \int_{-3}^{-3} f(t) \, dt = -6$$

$$g(2) = -6 + \int_{-3}^2 f(t) \, dt = -6 + 16 = 10$$

Since $g(-3) < 0 < g(2)$, by the Intermediate Value Theorem there is an x in the interval $[-3, 2]$ such that $g(x) = 0$.

6-6

Part B

For the second point: The response may also consider $x = 6$, though this is not required in order to justify an absolute minimum value or an absolute maximum value on the interval.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ considers $x = -5$, $x = -3$, $x = 2$ as candidates
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$g'(x) = f(x)$ changes sign only at $x = -5$, $x = -3$, and $x = 2$.

x	$g(x)$
-7	$-6 + \int_{-3}^{-7} f(t) dt = -6 - \int_{-7}^{-3} f(t) dt = -6 - (5 - 2) = -9$
-5	$-6 + \int_{-3}^{-5} f(t) dt = -6 - \int_{-5}^{-3} f(t) dt = -6 - (-2) = -4$
-3	-6
2	10
9	$-6 + \int_{-3}^9 f(t) dt = -6 + (16 - 5 - 6) = -1$

On the interval $-7 \leq x \leq 9$, the absolute minimum value is $g(-7) = -9$ and the absolute maximum value is $g(2) = 10$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the first point.

For the second point, the response must present a definite integral that correctly handles the substitution of the limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ expression for $\int (2x + 16) dx$
- ☐ integral in terms of u
- ☐ answer

Solution:

(i)

6-6

$$\int (2x + 16) \, dx = x^2 + 16x + C$$

(ii)

Let $u = 2x + 16$. Then $du = 2 \, dx$ and $\frac{1}{2} \, du = dx$.

$$x = -7 \Rightarrow u = 2(-7) + 16 = 2$$

$$x = -5 \Rightarrow u = 2(-5) + 16 = 6$$

$$\int_{-7}^{-5} f(2x + 16) \, dx = \frac{1}{2} \int_2^6 f(u) \, du = \frac{1}{2}(-5) = -\frac{5}{2}$$

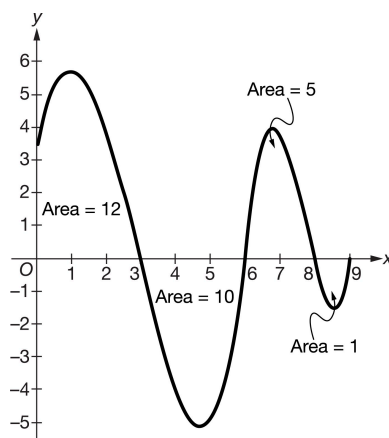
6-6

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The figure above shows the graph of f' , the derivative of a differentiable function f , on the closed interval $0 \leq x \leq 9$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(6) = 7$.

Let g be the function defined by $g(x) = x^2 - 1$.

- Find the value of $\int_0^9 f'(x) \, dx$.
- Given that $f(6) = 7$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of f and the absolute maximum value of f on the closed interval $0 \leq x \leq 9$. Justify your answers.
- Find $\int g(x) \, dx$.
- Find the value of $\int_2^3 x f'(g(x)) \, dx$.

Part A

The response should include supporting work for the numerical answer of 6.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^9 f'(x) \, dx = 12 - 10 + 5 - 1 = 6$$

Part B

The first point is earned for a correct expression for $f(x)$ OR evidence of correct use of FTC undefined in determining the appropriate $f(x)$ values for consideration.

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

6-6

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ use of Fundamental Theorem of Calculus
- ☐ considers $x = 3$, $x = 6$, and $x = 8$
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$$f(x) = f(6) + \int_6^x f'(t) dt$$

The absolute minimum and absolute maximum will occur at a critical point where $f'(x) = 0$

$$f'(x) = 0 \Rightarrow x = 3, x = 6, x = 8$$

The candidates are $x = 0$, $x = 3$, $x = 6$, $x = 8$, and $x = 9$.

x	$f(x)$
0	$f(6) + \int_6^0 f'(t) dt = f(6) - \int_0^6 f'(t) dt$ $= 7 - (12 - 10) = 5$
3	$f(6) + \int_6^3 f'(t) dt = f(6) - \int_3^6 f'(t) dt$ $= 7 - (-10) = 17$
6	7
8	$f(6) + \int_6^8 f'(t) dt = 7 + 5 = 12$
9	$f(6) + \int_6^9 f'(t) dt = 7 + (5 - 1) = 11$

On the interval $0 \leq x \leq 9$, the absolute minimum value is $f(0) = 5$ and the absolute maximum value is $f(3) = 17$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer.

Solution:

$$\int g(x) dx = \int (x^2 - 1) dx = \frac{1}{3}x^3 - x + C$$

Part D

To earn the second point, a response must present a definite integral that correctly handles the substitution of the limits of integration. Correct handling of the reversal of the limits of integration and all calculations are part of the third point.

6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student accurately includes all three of the criteria below.

- ☐ sets $u = x^2 - 1$
- ☐ integral in terms of u
- ☐ answer

Solution:

Let $u = g(x) = x^2 - 1$.

Then $du = 2x \, dx$ and $\frac{1}{2} du = x \, dx$.

$$x = 3 \Rightarrow u = 3^2 - 1 = 8$$

$$x = 2 \Rightarrow u = 2^2 - 1 = 3$$

$$\int_2^3 x f'(g(x)) \, dx = \frac{1}{2} \int_3^8 f'(u) \, du = \frac{1}{2}(-10 + 5) = -\frac{5}{2}$$

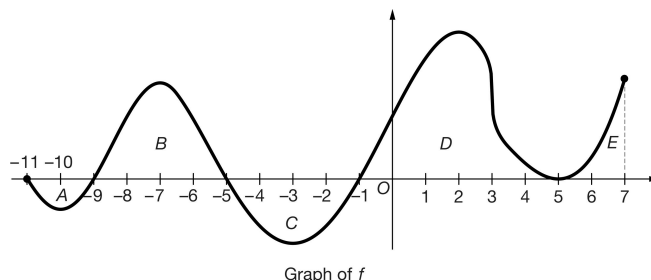
6-6

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The figure above shows the graph of the continuous function f . The regions A , B , C , D , and E have areas 2, 14, 10, 27, and 4, respectively. For $-11 \leq x \leq 7$, the function g is defined by $g(x) = 8 + \int_{-5}^x f(t) \, dt$.

- Is there a value of x , for $-5 \leq x \leq -1$, such that $g(x) = 0$? Justify your answer.
- Find the absolute minimum value of g and the absolute maximum value of g on the interval $-11 \leq x \leq 7$. Justify your answer.
- Find $\int (2x + 1) \, dx$.
 - Find the value of $\int_{-5}^{-1} f(2x + 1) \, dx$.

Part A

For the second point, the response must include that g is continuous on $-5 \leq x \leq -1$ show that $g(-1) < 0 < g(-5)$ and draw the conclusion from the Intermediate Value Theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g(-5)$ and $g(-1)$
- ☐ justification, using Intermediate Value Theorem

Solution:

$g'(x) = f(x)$ and f is continuous on $-11 \leq x \leq 7$. Because g is an antiderivative of a continuous function f , g is continuous on $-5 \leq x \leq -1$.

$$g(-5) = 8 + \int_{-5}^{-5} f(t) \, dt = 8$$

$$g(-1) = 8 + \int_{-5}^{-1} f(t) \, dt = 8 - 10 = -2$$

Since $g(-1) < 0 < g(-5)$, by the Intermediate Value Theorem there is an x in the interval $[-5, -1]$ such that $g(x) = 0$.

Part B

6-6

For the third and fourth points: At most 1 out of 2 points is earned for supporting work that justifies a correct absolute minimum value OR absolute maximum value based on a Candidates Test with a maximum of one arithmetic error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ considers $x = -9$, $x = -5$, $x = -1$ as candidates
- ☐ answers for absolute minimum value and absolute maximum value
- ☐ justification

Solution:

$g'(x) = f(x)$ changes sign only at $x = -9$, $x = -5$, and

x	$g(x)$
-11	$8 + \int_{-3}^{-11} f(t) dt = 8 - \int_{-11}^{-3} f(t) dt = 8 - (-2 + 14) = -4$
-9	$8 + \int_{-3}^{-9} f(t) dt = 8 - \int_{-9}^{-3} f(t) dt = 8 - 14 = -6$
-5	8
-1	-2
7	$8 + \int_{-3}^7 f(t) dt = 8 + (-10 + 27 + 4) = 29$

On the interval $-11 \leq x \leq 7$, the absolute minimum value is $g(-9) = -6$ and the absolute maximum value is $g(7) = 29$.

Part C

A response with any errors, including omission of the constant of integration, does not earn the first point.

For the second point, the response must present a definite integral that correctly handles the substitution of the limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ expression for $\int (2x + 1) dx$
- ☐ integral in terms of u
- ☐ answer

Solution:

(i)

$$\int (2x + 1) dx = x^2 + x + C$$

(ii)

Let $u = 2x + 1$. Then $du = 2 dx$ and $\frac{1}{2} du = dx$.

6-6

$$x = -1 \Rightarrow u = 2(-1) + 1 = -1$$

$$x = -5 \Rightarrow u = 2(-5) + 1 = -9$$

$$\int_{-5}^{-1} f(2x + 1) \, dx = \frac{1}{2} \int_{-9}^{-1} f(u) \, du = \frac{1}{2}(14 - 10) = 2$$

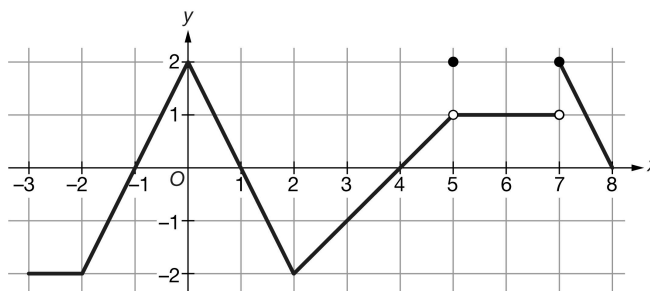
6-6

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the function f on the closed interval $-3 \leq x \leq 8$ consists of six line segments and the point $(5, 2)$, as shown in the figure above. The function g is given by $g(x) = \frac{1}{10}(4x^3 + 3x^2 - 10x - 17)$. It is known that $\int_{-3}^{-1} g(x) \, dx = -4.8$ and $\int_{-3}^4 g(x) \, dx = 11.2$.

- (a) Find the value of $\int_4^8 f(x) \, dx$, or explain why the integral does not exist.
- (b)
- (i) Find the value of $\int_{-1}^4 g(x) \, dx$. Show the work that leads to your answer.
- (ii) Find the value of $\int_{-1}^4 (2g(x) - 4f(x)) \, dx$. Show the work that leads to your answer.
- (c) Let $h(x) = \begin{cases} g(x) & \text{for } x \leq -1 \\ f(x) + b & \text{for } x > -1 \end{cases}$. Find the value of b for which $\int_{-3}^4 h(x) \, dx = 14.2$.
- (d) Find $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)+2}$. Show the work that leads to your answer.

Part A

The response should include supporting work for the numerical answer of 3.5.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The student response accurately includes the value of the integral.

Solution:

The integral exists since f is continuous on the interval $[4, 8]$, except for a removable discontinuity at $x = 5$ and a jump discontinuity at $x = 7$.

Using areas, $\int_4^8 f(x) \, dx = 0.5 + 2 + 1 = 3.5$.

Part B

Answers without supporting work do not earn any points.

6-6

A response that includes a sign error in an attempt to find $\int_{-1}^4 g(x) dx$ is eligible for the second and third points. A response that includes a sign error in an attempt to find $\int_{-1}^4 f(x) dx$ is eligible for the second point and not eligible for the third point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\int_{-1}^4 g(x) dx$
- ☐ uses integral properties
- ☐ answer

Solution:

$$(i) \int_{-1}^4 g(x) dx = \int_{-3}^4 g(x) dx - \int_{-3}^{-1} g(x) dx = 11.2 - (-4.8) = 16$$

$$(ii) \text{ Using areas, } \int_{-1}^4 f(x) dx = 2 - 3 = -1.$$

$$\begin{aligned} \int_{-1}^4 (2g(x) - 4f(x)) dx &= 2 \int_{-1}^4 g(x) dx - 4 \int_{-1}^4 f(x) dx \\ &= 2(16) - 4(-1) = 32 + 4 = 36 \end{aligned}$$

Part C

The evaluation of the definite integrals is assessed in the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses integral properties
- ☐ answer

Solution:

Using properties of integrals,

$$\begin{aligned} \int_{-3}^4 h(x) dx &= \int_{-3}^{-1} g(x) dx + \int_{-1}^4 (f(x) + b) dx \\ &= \int_{-3}^{-1} g(x) dx + \int_{-1}^4 f(x) dx + \int_{-1}^4 b dx \\ &= -4.8 + (-1) + (4 - (-1))b = -5.8 + 5b \\ -5.8 + 5b &= 14.2 \Rightarrow 5b = 20 \Rightarrow b = 4 \end{aligned}$$

Part D

For the first point, a response must show that $\lim_{x \rightarrow 1} f(x) = 0$ and $\lim_{x \rightarrow 1} (g(x) + 2) = 0$ and present an attempt at a limit of a ratio of derivatives. Use of $\frac{0}{0}$ in an equality statement with an expression or value is incorrect and results in the point not being earned.

6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ L'Hospital's Rule
- ☐ $g'(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 1} f(x) = 0 \text{ and } \lim_{x \rightarrow 1} (g(x) + 2) = -2 + 2 = 0.$$

Using L'Hospital's Rule,

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{f(x)}{g(x) + 2} &= \lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 1} \frac{f'(x)}{\frac{1}{10}(12x^2 + 6x - 10)} \\ &= \frac{-2}{\frac{1}{10}(12 + 6 - 10)} = \frac{-20}{8} = -\frac{5}{2}. \end{aligned}$$

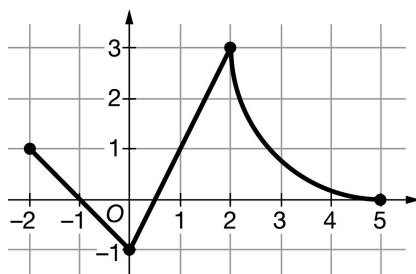
6-6

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $-6 \leq x \leq 5$. The figure above shows a portion of the graph of f , consisting of two line segments and a quarter of a circle centered at the point $(5, 3)$. It is known that the point $(3, 3 - \sqrt{5})$ is on the graph of f .

- If $\int_{-6}^5 f(x) \, dx = 7$, find the value of $\int_{-6}^{-2} f(x) \, dx$. Show the work that leads to your answer.
- Evaluate $\int_3^5 (2f'(x) + 4) \, dx$.
- The function g is given by $g(x) = \int_{-2}^x f(t) \, dt$. Find the absolute maximum value of g on the interval $-2 \leq x \leq 5$. Justify your answer.
- Find $\lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for showing evidence of using the property of adding integrals. This point can be earned symbolically or numerically with correct values of the integrals.

Simplification is not required to earn the second point or the third point

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_{-6}^5 f(x) \, dx = \int_{-6}^{-2} f(x) \, dx + \int_{-2}^5 f(x) \, dx$ integral
- ☐ $\int_{-2}^5 f(x) \, dx$
- ☐ answer

Solution:

6-6

$$\begin{aligned}\int_{-6}^5 f(x) dx &= \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx \\ \Rightarrow 7 &= \int_{-6}^{-2} f(x) dx + 2 + \left(9 - \frac{9\pi}{4}\right) \\ \Rightarrow \int_{-6}^{-2} f(x) dx &= 7 - \left(11 - \frac{9\pi}{4}\right) = \frac{9\pi}{4} - 4\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an antiderivative of $f'(x)$ and use of the limits of integration. Responses with errors in the integrand, such as a missing coefficient, earn the first point; such errors are considered for the second point.

Simplification is not required to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ answer

Solution:

$$\begin{aligned}\int_3^5 (2f'(x) + 4) dx &= 2 \int_3^5 f'(x) dx + \int_3^5 4 dx \\ &= 2(f(5) - f(3)) + 4(5 - 3) \\ &= 2\left(0 - \left(3 - \sqrt{5}\right)\right) + 8 \\ &= 2(-3 + \sqrt{5}) + 8 = 2 + 2\sqrt{5}\end{aligned}$$

OR

$$\begin{aligned}\int_3^5 (2f'(x) + 4) dx &= [2f(x) + 4x]_{x=3}^{x=5} \\ &= (2f(5) + 20) - (2f(3) + 12) \\ &= (2 \cdot 0 + 20) - \left(2\left(3 - \sqrt{5}\right) + 12\right) \\ &= 2 + 2\sqrt{5}\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for connecting $g'(x) = f(x)$ analytically or verbally.

The second point is earned with identification of $x = -1$ regardless of what other candidates are considered.

The third point is earned for calculating $g(-1)$ and $g(5)$, declaring the absolute maximum to be $11 - \frac{9\pi}{4}$, and correctly reasoning for excluding $g(-2)$. Note that the ordered pair $\left(5, 11 - \frac{9\pi}{4}\right)$ by itself is not a declaration of the absolute maximum.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$

6-6

- ☐ identifies $x = -1$ as a candidate
- ☐ answer with justification

Solution:

$$g'(x) = f(x) = 0 \Rightarrow x = -1, x = \frac{1}{2}, x = 5$$

x	$g(x)$
-2	0
-1	$\frac{1}{2}$
$\frac{1}{2}$	$-\frac{1}{4}$
5	$11 - \frac{9\pi}{4}$

On the interval $-2 \leq x \leq 5$, the absolute maximum value of g is $g(5) = 11 - \frac{9\pi}{4}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer. Simplification, including $\arctan 1 = \frac{\pi}{4}$, is not required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ answer

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x} &= \frac{10^1 - 3f'(1)}{f(1) - \arctan 1} \\ &= \frac{10 - 3 \cdot 2}{1 - \arctan 1} = \frac{4}{1 - \frac{\pi}{4}} \end{aligned}$$

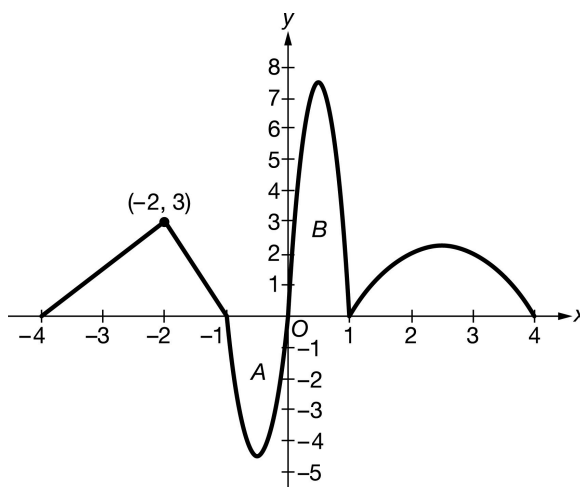
6-6

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined for $-4 \leq x \leq 4$. The graph of f , shown above, consists of two line segments and portions of three parabolas. The graph has horizontal tangents at $x = -\frac{1}{2}$, $x = \frac{1}{2}$, and $x = \frac{5}{2}$. It is known that $f(x) = -x^2 + 5x - 4$ for $1 \leq x \leq 4$. The areas of regions A and B bounded by the graph of f and the x -axis are 3 and 5, respectively. Let g be the function defined by $g(x) = \int_{-4}^x f(t) \, dt$.

- Find $g(0)$ and $g(4)$.
- Find the absolute minimum value of g on the closed interval $[-4, 4]$. Justify your answer.
- Find all intervals on which the graph of g is concave down. Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Simplification is not required for the first point.

The second point is earned with a correct integral or with work that shows that the integral has been taken. For example, $\int_1^4 f(t) \, dt$, or $\int_1^4 (-t^2 + 5t - 4) \, dt$, or $[-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t]_1^4$ would earn the second point.

The third point is earned with $-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t$.

Simplification is not required for the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ $g(0)$
- ☐ integral of f over $1 \leq t \leq 4$
- ☐ antiderivative

6-6

☐ $g(4)$

Solution:

$$g(0) = \int_{-4}^0 f(t) \, dt = \frac{9}{2} - 3 = \frac{3}{2}$$

$$\begin{aligned} g(4) &= \int_{-4}^4 f(t) \, dt \\ &= \int_{-4}^0 f(t) \, dt + \int_0^1 f(t) \, dt + \int_1^4 f(t) \, dt \\ &= \frac{3}{2} + 5 + \int_1^4 (-t^2 + 5t - 4) \, dt \\ &= \frac{3}{2} + 5 + \left[-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t \right]_1^4 \\ &= \frac{3}{2} + 5 + \left[\left(-\frac{1}{3} \cdot 4^3 + \frac{5}{2} \cdot 4^2 - 4 \cdot 4 \right) - \left(-\frac{1}{3} \cdot 1^3 + \frac{5}{2} \cdot 1^2 - 4 \cdot 1 \right) \right] \\ &= \frac{3}{2} + 5 + \left(\frac{8}{3} - \left(-\frac{11}{6} \right) \right) = 11 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an explicit statement connecting $g' = f$. Note that it is possible that this connection is made in an earlier part and will earn the point here.

The second point is earned for values appearing in an equation, table, or paragraph.

The third point is earned by ruling out all other candidates and committing to the minimum value of 0.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ identifies $x = -4$ and $x = 0$ as candidates
- ☐ answer with justification

Solution:

$g'(x) = f(x)$ is negative for $-1 < x < 0$, and nonnegative elsewhere. Thus, the absolute minimum value of g on $[-4, 4]$ can only occur at $x = -4$ or $x = 0$.

$$g(-4) = 0$$

$$g(0) = \frac{3}{2}$$

The absolute minimum value of g on $[-4, 4]$ is $g(-4) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with all three intervals, with or without the endpoints. Verbal expressions rather than intervals also earn the point.

The second point is earned with reasoning clearly tied to the given graph of f .



0	1	2
---	---	---

6-6

The student response accurately includes **both** of the criteria below.

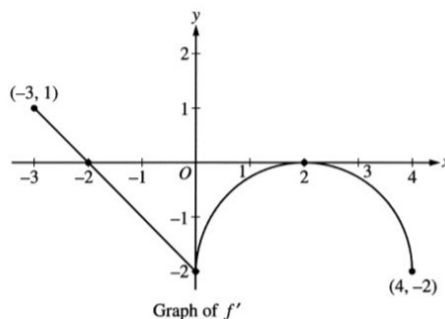
- ☐ intervals
- ☐ reason

Solution:

The graph of g is concave down on the intervals $-2 < x < -\frac{1}{2}$, $\frac{1}{2} < x < 1$, and $\frac{5}{2} < x < 4$ because $g'(x) = f(x)$ is decreasing on these intervals.

6-6

72.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

- On what intervals, if any, is f increasing? Justify your answer.
- Find the x -coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.
- Find an equation for the line tangent to the graph of f at the point $(0, 3)$.
- Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part A

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

Part C

6-6

1 point is earned for the correct equation

$$f(0) = -2$$

Tangent line is $y = -2x + 3$.

✓

0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct equation

$$f(0) = -2$$

Tangent line is $y = -2x + 3$.

Part D

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2 \right)$

$$\begin{aligned} f(0) - f(-3) &= \int_{-3}^0 f'(t) dt \\ &= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2} \end{aligned}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi \right)$

$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t) dt \\ &= - \left(8 - \frac{1}{2}(2)^2\pi \right) = -8 + 2\pi \end{aligned}$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2 \right)$

$$\begin{aligned} f(0) - f(-3) &= \int_{-3}^0 f'(t) dt \\ &= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2} \end{aligned}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

6-6

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi\right)$

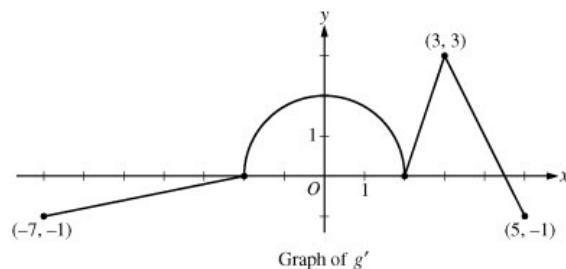
$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t) dt \\ &= -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

6-6

73.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

(a) Find $g(3)$ and $g(-2)$.

(b) Find the x -coordinate of each point of inflection of the graph of $y = g(x)$ on the interval $-7 < x < 5$. Explain your reasoning.

(c) The function h is defined by $h(x) = g(x) - \frac{1}{2}x^2$. Find the x -coordinate of each critical point of h , where $-7 < x < 5$, and classify each critical point as the location of a relative minimum, relative maximum, or neither a minimum nor a maximum. Explain your reasoning.

Part A

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$

Part B

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0$, $x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.

✓

0	1	2
---	---	---

6-6

The student response earns all of the following points:

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0$, $x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.

Part C

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}, 3$

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}, 3$

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

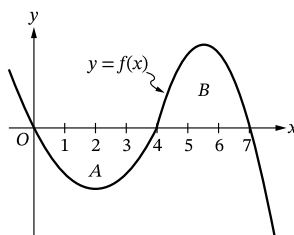
6-6

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.

74. Regions A and B in the figure shown are bounded by the graph of the function f and the x -axis.



If the area of each region is 3, what is the value of $\int_0^4 f(x)dx + \int_0^7 f(x)dx$?

- (A) -3
 (B) 0
 (C) 3
 (D) 9



Answer A

Correct. The definite integrals can be evaluated by using the connection between the definite integral and area.

$$\int_0^4 f(x)dx = -\text{Area}(A) = -3$$

$$\int_0^7 f(x)dx = -\text{Area}(A) + \text{Area}(B) = -3 + 3 = 0$$

$$\text{Therefore } \int_0^4 f(x)dx + \int_0^7 f(x)dx = -3 + 0 = -3.$$

75. What is the value of x at which the minimum value of $y = 3x^{\frac{4}{3}} - 2x$ occurs on the closed interval $[0, 1]$?

- (A) 0
 (B) $\frac{1}{8}$
 (C) $\frac{1}{2}$
 (D) 1



Answer B

Correct. The absolute minimum will occur at a critical point or one of the endpoints.

$$y' = 4x^{\frac{1}{3}} - 2 = 0 \Rightarrow x^{\frac{1}{3}} = \frac{1}{2} \Rightarrow x = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

The candidates are $x = 0$, $x = \frac{1}{8}$, and $x = 1$.

When $x = 0$, $y = 0$.

$$\text{When } x = \frac{1}{8}, y = 3\left(\frac{1}{8}\right)^{\frac{4}{3}} - 2\left(\frac{1}{8}\right) = \frac{3}{16} - \frac{1}{4} = -\frac{1}{16}.$$

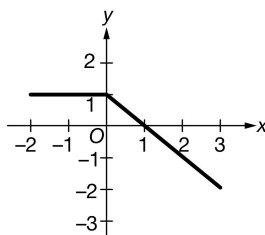
When $x = 1$, $y = 1$.

The absolute minimum is therefore at $x = \frac{1}{8}$.

Alternatively, since $x = \frac{1}{8}$ is the only critical point and the Second Derivative Test shows that it is the location of a local minimum, it must also be the location of the absolute minimum on the interval $[0, 1]$.

6-6

76.

Graph of f

The graph of the piecewise linear function f , which has a domain of $-2 \leq x \leq 3$, is shown in the figure above. What is the value of $\int_{-2}^3 f(x) \, dx$?

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{5}{2}$

(D) $\frac{9}{2}$

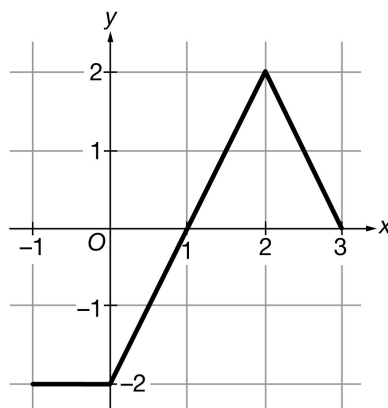
**Answer A**

Correct. The value of the definite integral can be found using geometry and the connection between the definite integral and area. Let area A be the rectangular region above the x -axis and to the left of the y -axis. Let area B be the triangular region above the x -axis and to the right of the y -axis. Let area C be the triangular region below the x -axis.

$$\begin{aligned}\int_{-2}^3 f(x) \, dx &= \text{Area}(A) + \text{Area}(B) - \text{Area}(C) \\ &= (2)(1) + \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) \\ &= 2 + \frac{1}{2} - 2 = \frac{1}{2}\end{aligned}$$

6-6

77.

Graph of f

The graph of the piecewise linear function f , which has a domain of $-1 \leq x \leq 3$, is shown in the figure above. What is the value of $\int_{-1}^3 f(x) \, dx$?

(A) -1 (B) 1 (C) 3 (D) 5 **Answer A**

Correct. The value of the definite integral can be found using geometry and the connection between the definite integral and area. Let area A be the rectangular region below the x -axis and to the left of the y -axis. Let area B be the triangular region below the x -axis and to the right of the y -axis. Let area C be the triangular region above the x -axis.

$$\begin{aligned} \int_{-1}^3 f(x) \, dx &= -\text{area}(A) - \text{area}(B) + \text{area}(C) \\ &= -(1)(2) - \frac{1}{2}(1)(2) + \frac{1}{2}(2)(2) = -1 \end{aligned}$$

78. If $\int_0^2 f(x) \, dx = 4$ and $\int_0^2 g(x) \, dx = -5$, then $\int_0^2 (3f(x) - g(x)) \, dx =$

(A) 7 (B) 9 (C) 17 (D) 27 **Answer C**

Correct.

$$\begin{aligned} \int_0^2 (3f(x) - g(x)) \, dx &= \int_0^2 3f(x) \, dx - \int_0^2 g(x) \, dx \\ &= 3 \int_0^2 f(x) \, dx - \int_0^2 g(x) \, dx = 3(4) - (-5) = 17 \end{aligned}$$

6-6

79.

$$f(x) = \begin{cases} 2 & \text{for } 0 \leq x < 2 \\ 3 & \text{for } 2 < x < 3 \\ -1 & \text{for } 3 \leq x \leq 8 \end{cases}$$

Let f be the piecewise function given above. The value of $\int_0^8 f(x) \, dx$ is

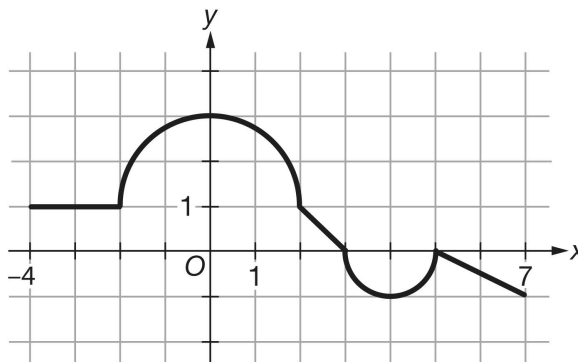
- (A) 2
(B) 12
(C) 32
(D) nonexistent

**Answer A**

Correct. Even though the function has a jump discontinuity at $x = 2$ and at $x = 3$, the value of the integral can be obtained by computing the definite integral in terms of areas of rectangles in the same way as a Riemann sum is constructed. Whether the rectangle is above or below the x -axis must also be taken into account.

$$\begin{aligned} \int_0^8 f(x) \, dx &= 2 \cdot (2 - 0) + 3 \cdot (3 - 2) + (-1) \cdot (8 - 3) \\ &= 2(2) + 3(1) + (-1)(5) = 2 \end{aligned}$$

80.

Graph of f

The graph of the function f on the interval $-4 \leq x \leq 7$ consists of three line segments and two semicircles, as shown in the figure above. What is the value of $\int_{-4}^7 f(x) \, dx$?

- (A) $\frac{3}{2}\pi + \frac{3}{2}$
(B) $\frac{3}{2}\pi + \frac{11}{2}$
(C) $\frac{5}{2}\pi + \frac{7}{2}$
(D) $\frac{5}{2}\pi + \frac{15}{2}$

**Answer B**

Correct. The definite integral can be evaluated by using geometry and the connection between the definite integral and area. Let A be the rectangular region on the interval $-4 < x < -2$. Let B and C be the rectangular and semicircular regions, respectively, on the interval $-2 < x < 2$. Let D be the triangular region on the interval $2 < x < 3$. Let E be the semicircular region on the interval $3 < x < 5$. Let F be the triangular region on the interval $5 < x < 7$.

$$\begin{aligned} \int_{-4}^7 f(x) \, dx &= \text{area}(A) + (\text{area}(B) + \text{area}(C)) \\ &\quad + \text{area}(D) - \text{area}(E) - \text{area}(F) \\ &= 2 + \left(4 + \frac{1}{2}(2)^2\pi\right) + \frac{1}{2} - \frac{1}{2}(1)^2\pi - \frac{1}{2}(2)(1) = \frac{3}{2}\pi + \frac{11}{2} \end{aligned}$$

6-6

81. If $\int_{-1}^3 (2g(x) + 4) dx = 22$ and $\int_{10}^{-1} g(x) dx = 12$, then $\int_3^{10} g(x) dx =$
- (A) -21
- (B) -15
- (C) -9
- (D) 9

Answer B

Correct. Using the properties of definite integrals, $22 = \int_{-1}^3 (2g(x) + 4) dx = \int_{-1}^3 2g(x) dx + \int_{-1}^3 4 dx = 2 \int_{-1}^3 g(x) dx + 16 \Rightarrow \int_{-1}^3 g(x) dx = 3$.

Also, $\int_{-1}^{10} g(x) dx = -\int_{10}^{-1} g(x) dx = -12$. Since $\int_{-1}^3 g(x) dx + \int_3^{10} g(x) dx = \int_{-1}^{10} g(x) dx$,

$$\int_3^{10} g(x) dx = \int_{-1}^{10} g(x) dx - \int_{-1}^3 g(x) dx = -12 - 3 = -15.$$

82. If $f(x) = \frac{(x-1)(x+2)}{|x-1|}$, then the value of $\int_{-1}^3 f(x) dx$ is
- (A) -4
- (B) 4
- (C) 12
- (D) nonexistent

Answer B

Correct. This function has a jump discontinuity at $x = 1$. $f(x) = -(x + 2)$ for $x < 1$ and $f(x) = x + 2$ for $x > 1$. Consider the graph of f and let region A be between the graph of f and the x -axis on the interval $[-1, 1]$, and let region B be between the graph of f and the x -axis on the interval $[1, 3]$. The definite integral over the interval $[-1, 3]$ can be written as the sum of the definite integrals over $[-1, 1]$ and $[1, 3]$. The value of the integral can be obtained by computing the definite integral using areas and taking into account which region is below the x -axis and which region is above the x -axis.

$$\int_{-1}^3 f(x) dx = -\text{area}(A) + \text{area}(B) = -2\left(\frac{1+3}{2}\right) + 2\left(\frac{3+5}{2}\right) = -4 + 8 = 4$$

83. If $\int_{-1}^1 f(x) dx = 7$ and $\int_{-1}^1 g(x) dx = -9$, then $\int_{-1}^1 (2f(x) - g(x)) dx =$
- (A) 5
- (B) 16
- (C) 23
- (D) 32

Answer C

Correct.

$$\begin{aligned} \int_{-1}^1 (2f(x) - g(x)) dx &= \int_{-1}^1 2f(x) dx - \int_{-1}^1 g(x) dx \\ &= 2 \int_{-1}^1 f(x) dx - \int_{-1}^1 g(x) dx = 2(7) - (-9) = 14 + 9 = 23 \end{aligned}$$

6-6

84.

$$f(x) = \begin{cases} -1 & \text{for } 0 \leq x < 3 \\ 1 & \text{for } 3 \leq x < 5 \\ 3 & \text{for } 5 \leq x \leq 7 \end{cases}$$

Let f be the piecewise function given above. The value of $\int_0^7 f(x) \, dx$ is

(A) 5

(B) 11

(C) 35

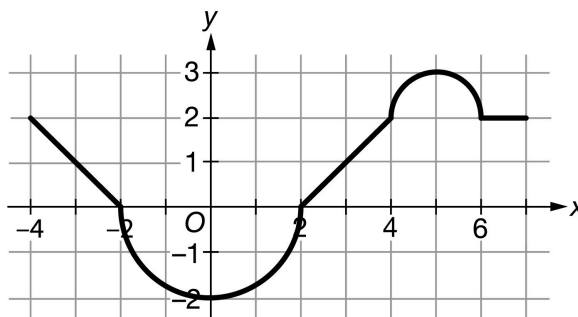
(D) nonexistent

**Answer A**

Correct. Even though the function has a jump discontinuity at $x = 3$ and at $x = 5$, the value of the integral can be obtained by computing the definite integral in terms of areas of rectangles in the same way as a Riemann sum is constructed. Whether the rectangle is above or below the x -axis must also be taken into account.

$$\begin{aligned} \int_0^7 f(x) \, dx &= (3 - 0)(-1) + (5 - 3)(1) + (7 - 5)(3) \\ &= -3 + 2 + 6 = 5 \end{aligned}$$

85.

Graph of f

The graph of the function f on the interval $-4 \leq x \leq 7$ consists of three line segments and two semicircles, as shown in the figure above. What is the value of $\int_{-4}^7 f(x) \, dx$?

(A) $6 - \frac{3\pi}{2}$ (B) $10 - \frac{3\pi}{2}$ (C) $6 + \frac{5\pi}{2}$ (D) $10 + \frac{5\pi}{2}$ **Answer B**

Correct. The definite integral can be evaluated by using geometry and the connection between the definite integral and area. Let A be the triangular region on the interval $-4 < x < -2$. Let B be the semicircular region on the interval $-2 < x < 2$. Let C be the triangular region on the interval $2 < x < 4$. Let D be the rectangular and semicircular region on the interval $4 < x < 6$. Let E be the rectangular region on the interval $6 < x < 7$.

$$\begin{aligned} \int_{-4}^7 f(x) \, dx &= \text{Area}(A) - \text{Area}(B) + \text{Area}(C) \\ &\quad + \text{Area}(D) + \text{Area}(E) \\ \int_{-4}^7 f(x) \, dx &= \frac{1}{2}(2)(2) - \frac{\pi(2^2)}{2} + \frac{1}{2}(2)(2) + \frac{\pi(1)^2}{2} + (2)(2) + (1)(2) = 2 - 2\pi + 2 + \frac{\pi}{2} + 4 + 2 = 10 - \frac{3\pi}{2} \end{aligned}$$

6-6

86. If $\int_{-2}^8 (3g(x) + 2) \, dx = 35$ and $\int_5^{-2} g(x) \, dx = -12$, then $\int_5^8 g(x) \, dx =$
- (A) -23
- (B) -7
- (C) 19
- (D) 23

Answer B

Correct. Using the properties of definite integrals,

$$35 = \int_{-2}^8 (3g(x) + 2) \, dx = \int_{-2}^8 3g(x) \, dx + \int_{-2}^8 2 \, dx = 3 \int_{-2}^8 g(x) \, dx + 2x \Big|_{-2}^8 = 3 \int_{-2}^8 g(x) \, dx + 2(8 - (-2)) = 3 \int_{-2}^8 g(x) \, dx + 20.$$

Then, $\int_{-2}^8 g(x) \, dx = \frac{35 - 20}{3} = 5$ and $\int_5^{-2} g(x) \, dx = - \int_{-2}^5 g(x) \, dx$.

Also, $\int_{-2}^8 g(x) \, dx = \int_{-2}^5 g(x) \, dx + \int_5^8 g(x) \, dx$. That is, $5 = -(-12) + \int_5^8 g(x) \, dx$.

Therefore, $\int_5^8 g(x) \, dx = 5 - 12 = -7$.

87. If $f(x) = \frac{|x|(x^2-1)}{x}$, then the value of $\int_{-1}^2 f(x) \, dx$ is
- (A) 0
- (B) $\frac{4}{3}$
- (C) $\frac{9}{4}$
- (D) nonexistent

Answer B

Correct. This function has a jump discontinuity at $x = 0$. $f(x) = -(x^2 - 1)$ for $x < 0$ and $f(x) = x^2 - 1$ for $x > 0$. The definite integral $\int_{-1}^2 f(x) \, dx$ can be

written as $\int_{-1}^2 f(x) \, dx = \int_{-1}^0 -(x^2 - 1) \, dx + \int_0^2 (x^2 - 1) \, dx$.

Since $\int_{-1}^0 -(x^2 - 1) \, dx = -\frac{x^3}{3} + x \Big|_{-1}^0 = 0 - \left(\frac{1}{3} - 1\right) = \frac{2}{3}$

and

$$\int_0^2 (x^2 - 1) \, dx = \frac{x^3}{3} - x \Big|_0^2 = \left(\frac{8}{3} - 2\right) - 0 = \frac{2}{3},$$

therefore, $\int_{-1}^2 f(x) \, dx = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$.

6-6

88. $\int_{-1}^3 \frac{|x|}{x} dx$ is

(A) $\frac{5}{4}$

(B) 2

(C) 4

(D) undefined



Answer B

Correct. The function $\frac{|x|}{x}$ has a jump discontinuity at $x = 0$. The definition of the definite integral may be extended to functions with jump discontinuities, and the integral in this case can be computed using geometry. The integrand has the value -1 on the interval $[-1, 0)$ of length 1 and has the value $+1$ on the interval $(0, 3]$ of length 3. Therefore, $\int_{-1}^3 \frac{|x|}{x} dx = (1)(-1) + 3(1) = -1 + 3 = 2$.

89. Let h be a continuous function such that $\int_3^7 h(x) dx = -6$. What is the value of $\int_7^3 (5h(x) + 2) dx$?

(A) 38

(B) 22

(C) -28

(D) -38



Answer B

Correct. Using the properties of definite integrals,

$$\int_7^3 (5h(x) + 2) dx = \int_7^3 5h(x) dx + \int_7^3 2 dx = -5 \int_3^7 h(x) dx - \int_3^7 2 dx = -5(-6) - (7 - 3)(2) = 30 - 8 = 22$$

90. Let f be a continuous function such that $f(x) = f(-x)$ for all x . If $\int_{-2}^0 f(x) dx = 4$ and $\int_1^2 f(x) dx = 1$, what is the value of $\int_0^1 f(x) dx$?

(A) -5

(B) 2

(C) 3

(D) 5



Answer C

Correct. By the property of definite integrals, $\int_0^1 f(x) dx + \int_1^2 f(x) dx = \int_0^2 f(x) dx \Rightarrow \int_0^1 f(x) dx = \int_0^2 f(x) dx - \int_1^2 f(x) dx$.

The graph of f is symmetric with respect to the y -axis, because $f(x) = f(-x)$. Therefore, $\int_0^2 f(x) dx = \int_{-2}^0 f(x) dx = 4$.

Then $\int_0^1 f(x) dx = \int_0^2 f(x) dx - \int_1^2 f(x) dx = 4 - 1 = 3$.

6-6

91. Let g be a continuous function such that $\int_a^b 2g(x) \, dx = 5$ and $\int_a^c 4g(x) \, dx = 7$. What is the value of $\int_b^c g(x) \, dx$?

- (A) $-\frac{3}{4}$
(B) 2
(C) $\frac{17}{4}$
(D) 18

**Answer A**

Correct. The properties of the definite integral are used as follows.

$$5 = \int_a^b 2g(x) \, dx = 2 \int_a^b g(x) \, dx \Rightarrow \int_a^b g(x) \, dx = \frac{5}{2}$$

$$7 = \int_a^c 4g(x) \, dx = 4 \int_a^c g(x) \, dx \Rightarrow \int_a^c g(x) \, dx = \frac{7}{4}$$

$$\int_a^b g(x) \, dx + \int_b^c g(x) \, dx = \int_a^c g(x) \, dx \Rightarrow \int_b^c g(x) \, dx = \int_a^c g(x) \, dx - \int_a^b g(x) \, dx = \frac{7}{4} - \frac{5}{2} = -\frac{3}{4}$$

92. Let f be a continuous function. If $\int_4^5 f(x) \, dx = 20$, $\int_5^k f(x) \, dx = -5$, and $\int_4^k (f(x) + 2) \, dx = 45$, what is the value of k ?

- (A) 13
(B) 15
(C) 17
(D) 19

**Answer D**

Correct. Using properties of definite integrals gives $\int_4^k f(x) \, dx = \int_4^5 f(x) \, dx + \int_5^k f(x) \, dx = 20 + (-5) = 15$ and

$$45 = \int_4^k (f(x) + 2) \, dx = \int_4^k f(x) \, dx + \int_4^k 2 \, dx = 15 + 2(k - 4) = 7 + 2k. \text{ Then } 2k = 45 - 7 = 38 \Rightarrow k = 19.$$

6-6

93. If $\int_1^4 f(x) \, dx = 8$ and $\int_1^4 g(x) \, dx = -2$, which of the following cannot be determined from the information given?

(A) $\int_4^1 g(x) \, dx$

(B) $\int_1^4 3f(x) \, dx$

(C) $\int_1^4 3f(x)g(x) \, dx$

(D) $\int_1^4 (3f(x) + g(x)) \, dx$

**Answer C**

Correct. It is not true in general that $\int_1^4 3f(x)g(x) \, dx = \int_1^4 3f(x) \, dx \cdot \int_1^4 g(x) \, dx$, so the individual values of $\int_1^4 f(x) \, dx$ and $\int_1^4 g(x) \, dx$ cannot be used to determine the value of $\int_1^4 3f(x)g(x) \, dx$. For example, if $f(x) = \frac{8}{3}$ and $g(x) = -\frac{2}{3}$, then $\int_1^4 f(x) \, dx = 8$, $\int_1^4 g(x) \, dx = -2$, and $\int_1^4 3f(x)g(x) \, dx = \int_1^4 -\frac{16}{3} \, dx = -16$. However, if $f(x) = \frac{16}{9}(x-1)$ and $g(x) = -\frac{4}{9}(x-1)$, then $\int_1^4 f(x) \, dx = 8$ and $\int_1^4 g(x) \, dx = -2$ as before, but now $\int_1^4 3f(x)g(x) \, dx = \int_1^4 -\frac{64}{27}(x-1)^2 \, dx = -\frac{64}{3}$.

94. If $\int_a^b 2f(x) \, dx + \int_c^b f(x) \, dx = 5$ and $\int_a^c f(x) \, dx = 1$, then $\int_a^b f(x) \, dx =$

(A) 4

(B) 2

(C) 1

(D) -1

**Answer B**

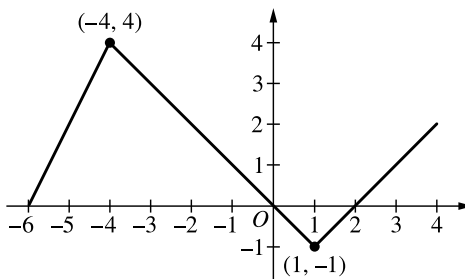
Correct. Using properties of the definite integral gives the following.

$$5 + 1 = \left(\int_a^b 2f(x) \, dx + \int_c^b f(x) \, dx \right) + \int_a^c f(x) \, dx = 2 \int_a^b f(x) \, dx + \left(\int_c^b f(x) \, dx + \int_a^c f(x) \, dx \right) = 2 \int_a^b f(x) \, dx + \int_a^b f(x) \, dx = 3 \int_a^b f(x) \, dx$$

Therefore $\int_a^b f(x) \, dx = \frac{6}{3} = 2$.

6-6

95.

Graph of f

The graph of a piecewise linear function f is shown in the figure. If $g(x) = \int_2^x f(t) \, dt$, what is the value of $g(-6)$?

- (A) -13
 (B) -11
 (C) 11
 (D) 13

**Answer B**

Correct. The definite integral can be evaluated by using properties of integrals as well as geometry and the connection between the definite integral and area.

$$g(-6) = \int_2^{-6} f(t) \, dt = - \int_{-6}^2 f(t) \, dt = - \left(\frac{1}{2}(6)(4) - \frac{1}{2}(2)(1) \right) = -(12 - 1) = -11$$

96. A particle moves along a straight line so that at time $t \geq 0$ its acceleration is given by $a(t) = 12t$. At time $t = 0$, the velocity of the particle is 2 and the position of the particle is 5.

Which of the following is an expression for the position of the particle at time $t \geq 0$?

- (A) $6t^2 + 5$
 (B) $6t^3 + 2t + 5$
 (C) $2t^3 + 5$
 (D) $2t^3 + 2t + 5$

**Answer D**

Correct. Since the acceleration is given, the position can be found using antidifferentiation and the values of the velocity and position at time $t = 0$.

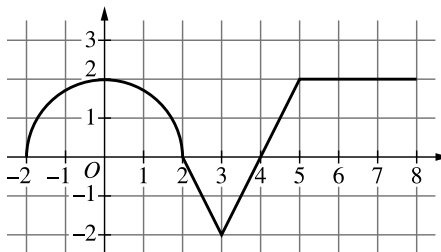
$$a(t) = 12t \Rightarrow v(t) = 6t^2 + C_1; v(0) = 2 \Rightarrow 2 = 0 + C_1 \Rightarrow C_1 = 2$$

$$v(t) = 6t^2 + 2 \Rightarrow s(t) = 2t^3 + 2t + C_2; s(0) = 5 \Rightarrow 5 = 0 + 0 + C_2 \Rightarrow C_2 = 5$$

The position of the particle is $s(t) = 2t^3 + 2t + 5$ for $t \geq 0$.

6-6

97.

Graph of f

The graph of the function f shown consists of a semicircle centered at the origin and three line segments.

If $g(x) = \int_{-2}^x (f(t) + 1) \, dt$, then $g(8) =$

- (A) $2\pi + 6$
(B) $2\pi + 10$
(C) $2\pi + 15$
(D) $2\pi + 19$

**Answer C**

Correct. $g(8) = \int_{-2}^8 (f(t) + 1) \, dt$. The definite integral can be evaluated by using properties of integrals as well as geometry and the connection between the definite integral and area.

$$g(8) = \int_{-2}^8 (f(t) + 1) \, dt = \int_{-2}^8 f(t) \, dt + \int_{-2}^8 1 \, dt = \left(\frac{1}{2}\pi(2)^2 - \frac{1}{2}(2)(2) + \frac{1}{2}(1)(2) + (3)(2) \right) + (8 - (-2)) = (2\pi - 2 + 1 + 6) + 10 = 2\pi + 15$$